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CHAPTER

ATA 49

AUXILIARY POWER UNIT

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Auxiliary Power Unit (APU) - General

Introduction

The APS 1000–C12 Auxiliary Power Unit (APU) is a turbine–powered system used to give auxiliary electrical power and pneumatic power.

General Description

The APU engine and its associated components are installed in the aft section of the tail cone. Consisting of a gas turbine engine driving a starter/generator, the APU gives bleed air to the air conditioning pack and 28 Vdc to the left main feeder bus. The APU is installed for ground use only and cannot be operated in flight.

The APU has a control system, an air induction system, an exhaust system, a gearbox, accessories, and an electrical generator. The APU includes the following:

- Powerplant
- Engine
- Engine Fuel and Control
- Ignition and Starting
- Air
- Engine Controls
- Indicating
- Oil

Leading Particulars	
Weight	140 lbs (dry)
Operating Altitude	—1,000 to 15,000 ft (Ground Mode Only)
Sea Level 59 °F (15 °C) Standard Day:	
Fuel Consumption	202 lb/hr (20 Shaft horsepower 84lb/ min Bleed Flow)
Output Shaft Horsepower	60 HP
Rotor Speed	100% (64,154 PRM)
Rotor Overspeed	105% (67,362 RPM)
Rotor Underspeed	95% (54,531 RPM)
Exhaust Gas Temperature (EDT):	
Start Acceleration	1,890 °F (1,032 °C)
Steady State	1,324 °F (717 °C)
Oil Pressure	35 PSIG (minimum) 60 PSIG (maximum)
Oil Temperature	275 °C (135 °C)(maximum)
Oil Type	MIL-L-7808 (-54 to +54 °C (-65 to +130 °F)) MIL-L-23699 (-40 to +54 °C (-40 to +130 °F))

Table 49-1 Leading Particulars

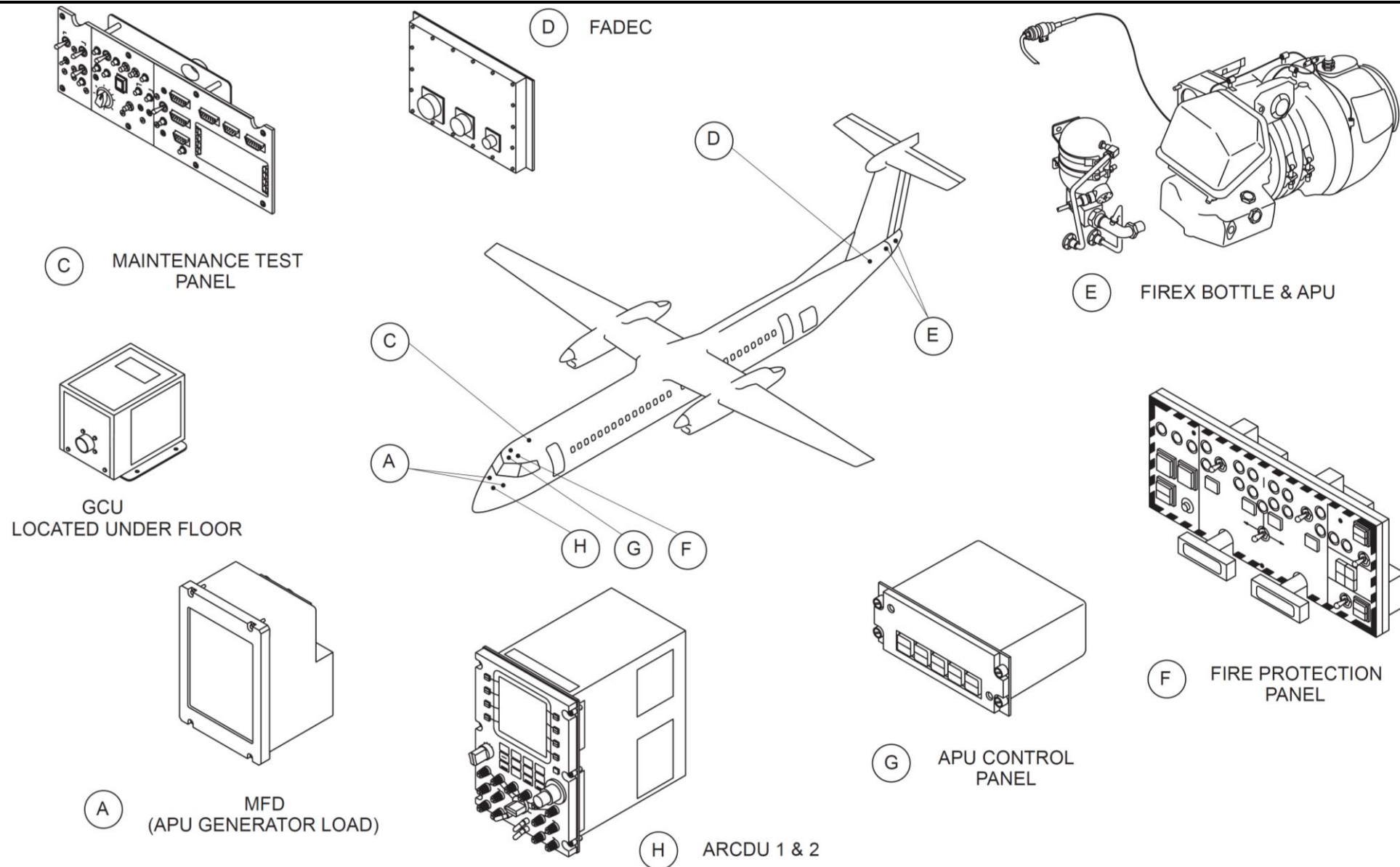


Figure 49-1 APU & Related Components

Detailed Description

[Refer Figure 49-1 & 49-2](#)

The APU control consists of a Full Authority Digital Electronic Controller (FADEC) and various valves, and sensors. The FADEC controls, monitors and diagnoses all phases of the APU system operation through these valves and sensors. The FADEC also communicates with the aircraft system through an ARINC link. The FADEC controls the APU start, acceleration to 100% speed, the starter motor, the exciter and fuel flow. The FADEC monitors engine speed, EGT, oil temperature, oil pressure, and inlet pressure.

During operation, the APU rotor speed is kept at a constant speed. Abnormal conditions during operation are recorded as faults in the FADEC Non Volatile Memory. The FADEC provides the necessary hardware and software for control sequencing, condition monitoring, fault detection, fault isolation and protection of the APU System. The FADEC also performs the interface functions between the aircraft and the APU.

The air induction system consists of an air inlet system located in the aircraft tail cone. The air induction system provides outside air to the APU compressor, which compresses the air. Some of the compressed air is bled off the APU and delivered to the aircraft Environmental Control System. A bleed control valve controls the flow of air.

The remaining compressed air is heated in a combustor using fuel provided by the APU fuel system, the combustion products then drive the turbine wheel. The energy released drives the compressor and the gearbox with its mounted accessories. The exhaust gases exit the APU through the exhaust system.

The gearbox mounted electrical generator provides electrical power used by the aircraft systems. The gearbox also drives the following accessories: an oil pump for lubrication, a fuel control which provides fuel to the combustor.

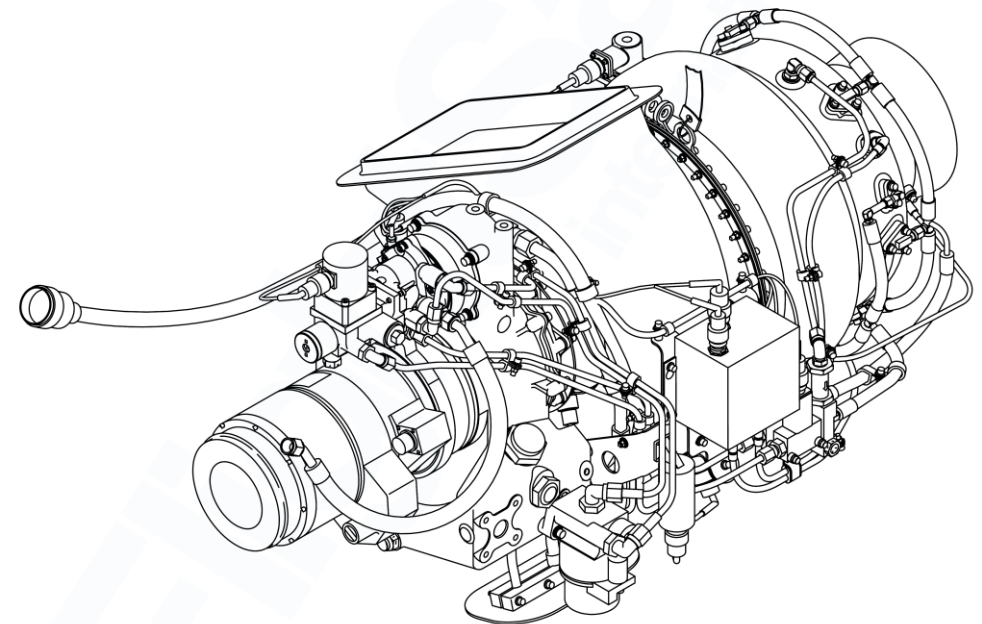
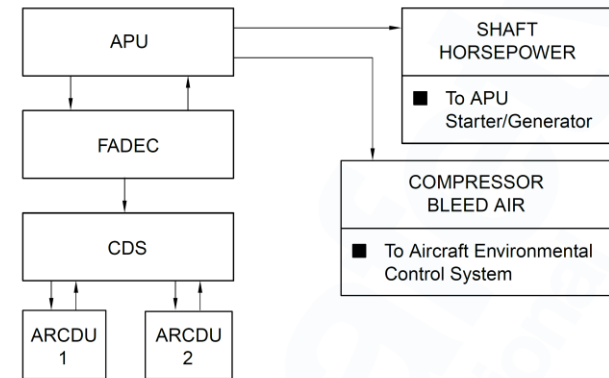


Figure 49-2 APU Synoptic

Power Plant

Introduction

The Auxiliary Power Unit gives electrical power to the aircraft left main bus, and air conditioning to the Environmental Control System, when the aircraft is in the ground. The APU powerplant is installed in a compartment that provides interfaces for the APU fuel line, electrical connections, air inlet assembly, exhaust, starter/generator duct system, fire detection and the fire extinguishing system. –

General Description

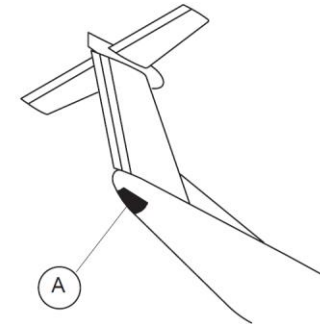
[Refer Figures 49-3 & 49-4](#)

The Auxiliary Power Unit (APU) is located in a fireproof titanium compartment in the tail cone of the aircraft. Access to the APU is through two hinged doors that are part of the tail cone structure.

The APU is installed to the tail cone structure by three mounts. The two forward mounts are attached to the gearbox mounts and the rear strut is attached to the APU rear mount. A fish pole hoist is used to install and remove the APU. The hoist is attached to a lug in the APU compartment and uses quick connects to the APU front and rear lifting brackets.

The APU includes the components that follow:

- Harness, Electrical-APU
- Harness, Electrical-Interconnect
- Mounts, Side
- Mounts, Aft
- Duct, Inlet
- Drain, Exhaust
- Vent, Gearbox
- Drain, Fuel Pump
- Drain, Inlet
- Drain, Combustor



LEGEND

1. Latch, Right Access Door.
2. Latch, Left Access Door.
3. Drain Mast.
4. Starter/Generator Intake.
5. Tailcone Ventilation Intake.

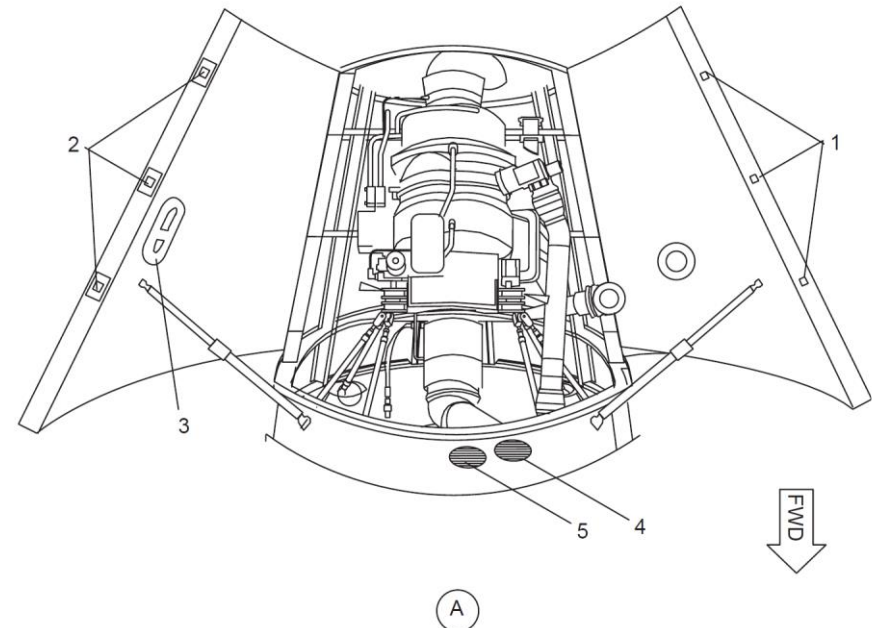


Figure 49-3 APU Access & Lower Intakes

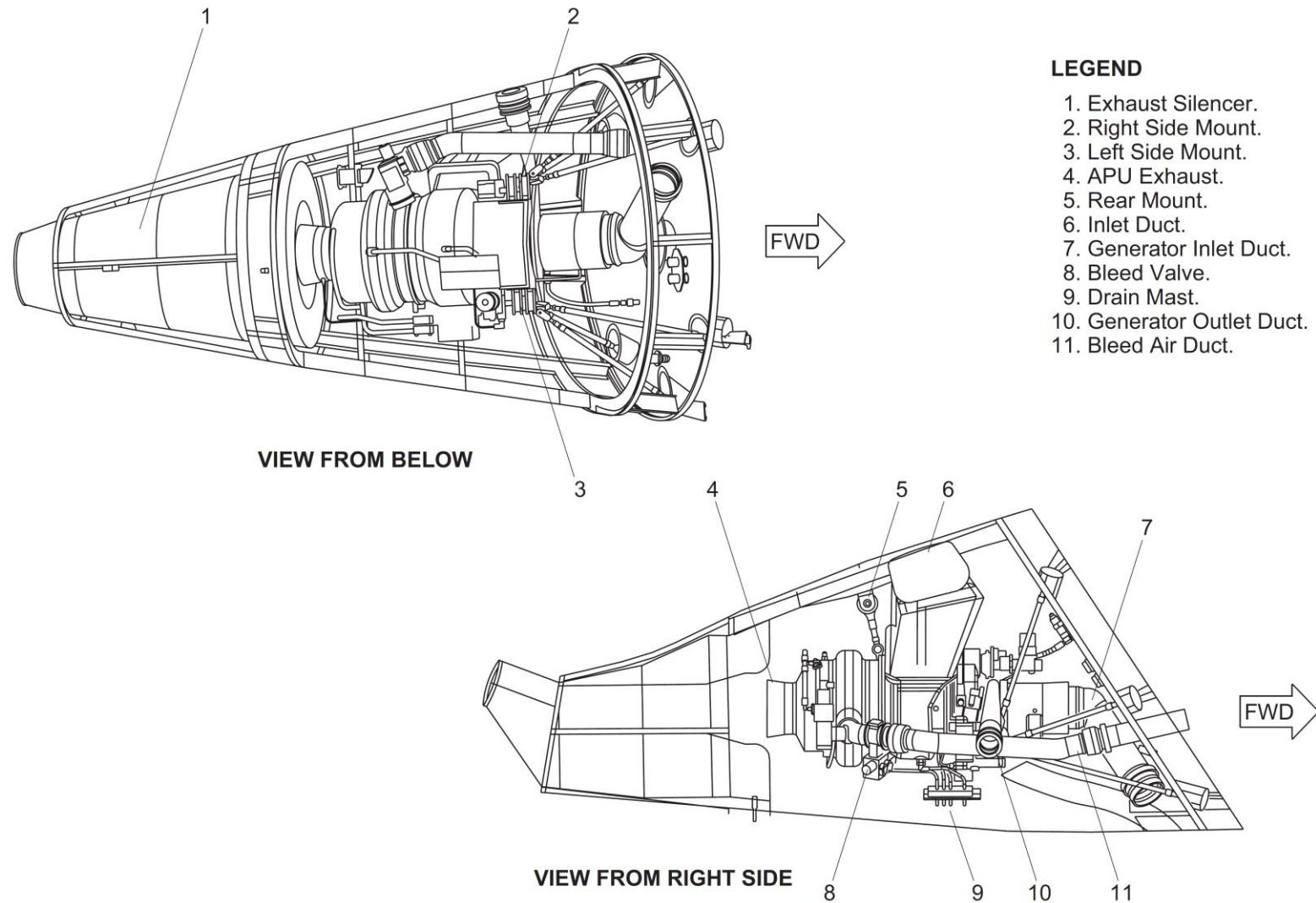


Figure 49-4 APU Powerplant View

Detailed Description

The Auxiliary Power Unit mounting system is designed to be failsafe. Limit flight loads will be supported with any single component failure.

The mounting system includes the components that follow:

- Six main struts
- Two gearbox mounts
- One aft mount

Power Plant – Component Description

Harness, Electrical–APU

[Refer Figure 49-5A](#)

The engine control harness assembly consists of 15 electrical connectors and associated connecting wires. When installed on the APU engine assembly, the engine control harness assembly gives the electrical interface between the electrical components and the connection at the tail cone firewall bulkhead.

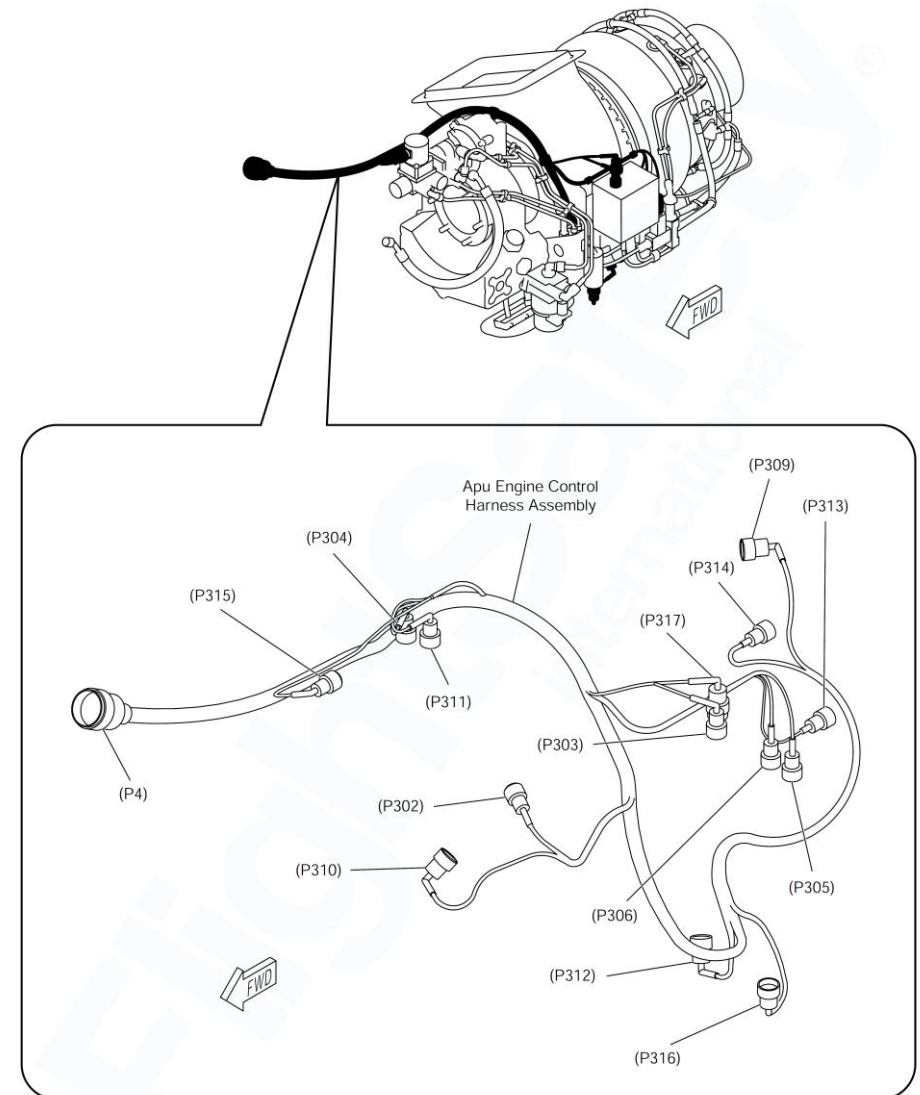


Figure 49-5A APU Harness

Harness, Electrical-Interconnect

[Refer Figure 49-5B](#)

The electrical inter-connect harness carries the electrical interface from the connection point on the tail cone firewall bulkhead, to the APU Full Authority Digital Electronic Control (FADEC) which is installed in the air conditioning bay in the tail cone.

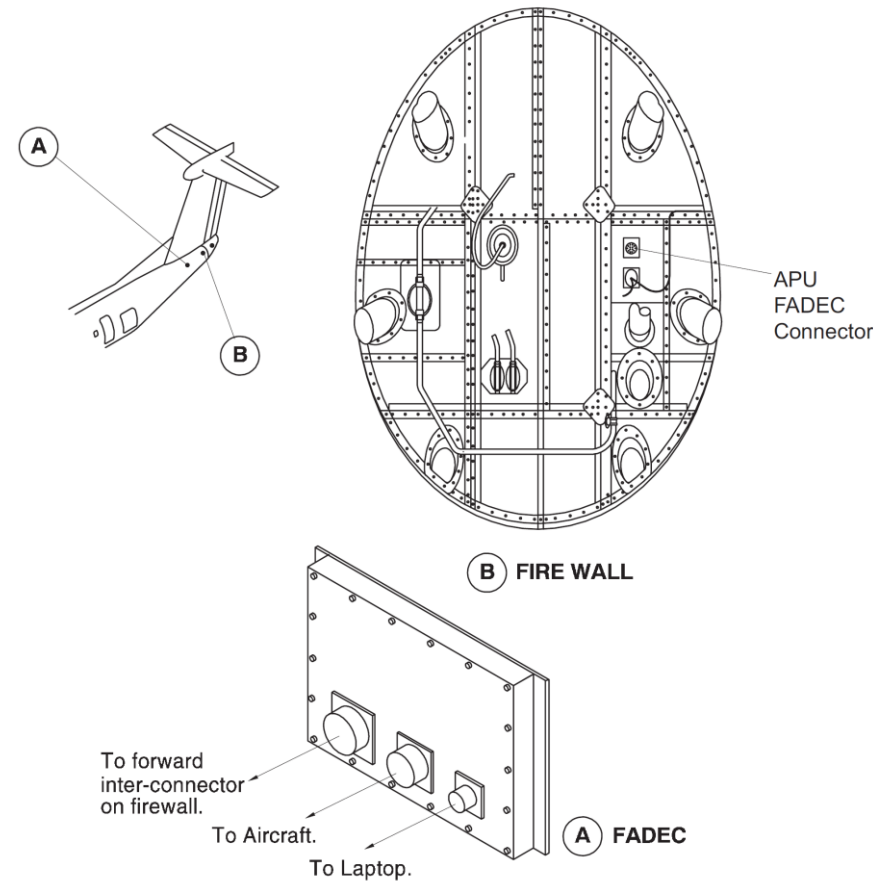


Figure 49-5B APU FADEC Connectors

Mounts, Side

[Refer Figure 49-6](#)

The two APU side mounts provide two, of the three vibration isolated, failsafe supports for the APU in the tail cone structure. Each mount has three strut assemblies in a tripod arrangement, a strut bracket, two part vibration isolator, gearbox isolator mount and isolator snubbing washer.

Mount, Aft

[Refer Figure 49-6](#)

The aft APU mount is the third, of the three, vibration isolated, failsafe supports for the APU. The lower end of the aft mount is attached to the APU at the uppermost point of the turbine combustor flange. The upper end of the aft mount is attached to the lower side of the centre upper beam in the tail cone structure.

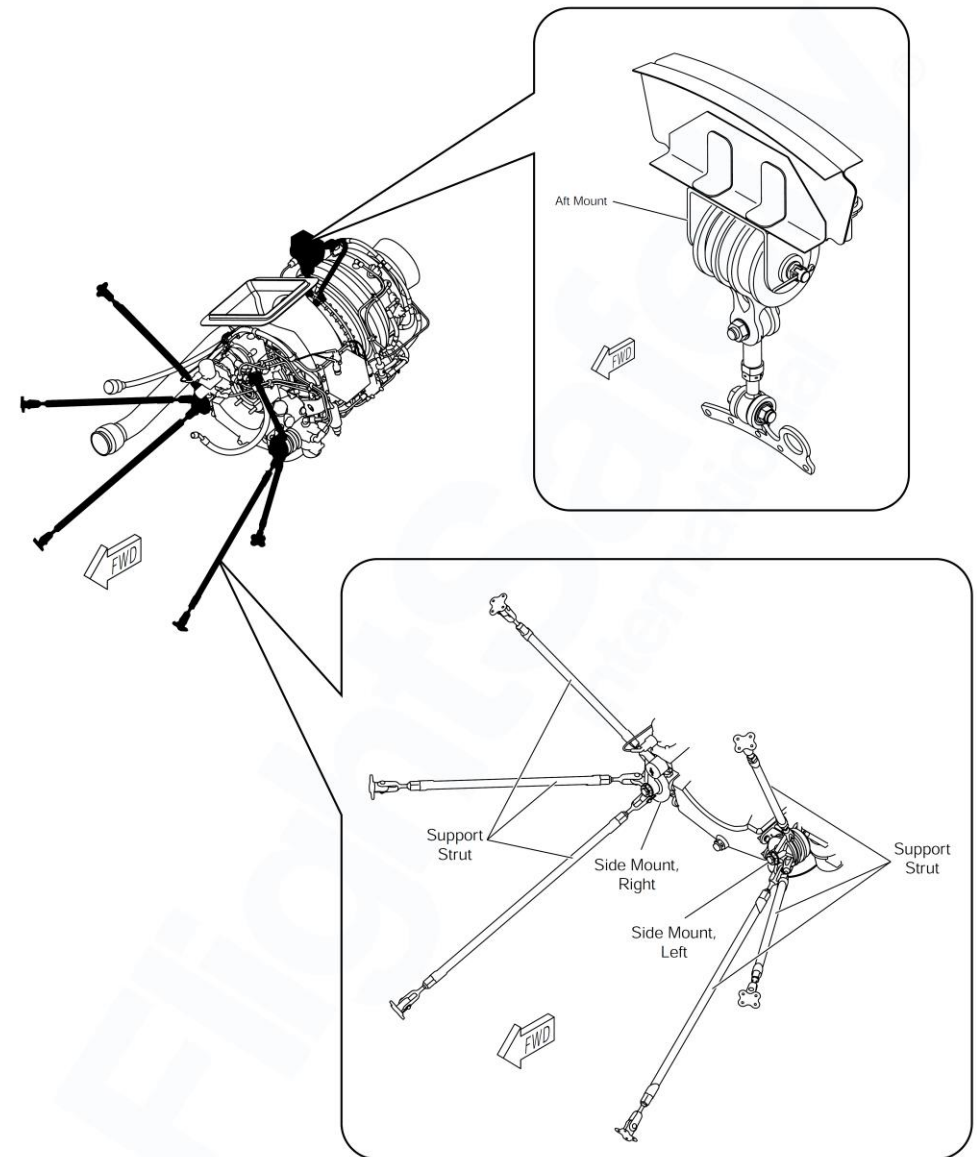


Figure 49-6 APU Front & Top Mounts

Duct, Inlet

[Refer Figure 49-7](#)

The duct is a fireproof, acoustically treated, bonded assembly.

There is a fireproof silicone rubber seal where the duct attaches to the APU.
This seal prevents tail cone air from entering the APU inlet system.

It also allows for minor misalignment and motion between the APU and the structure.

Inlet is protected by a mesh screen, which can be replaced by a louvered door for winter operations.

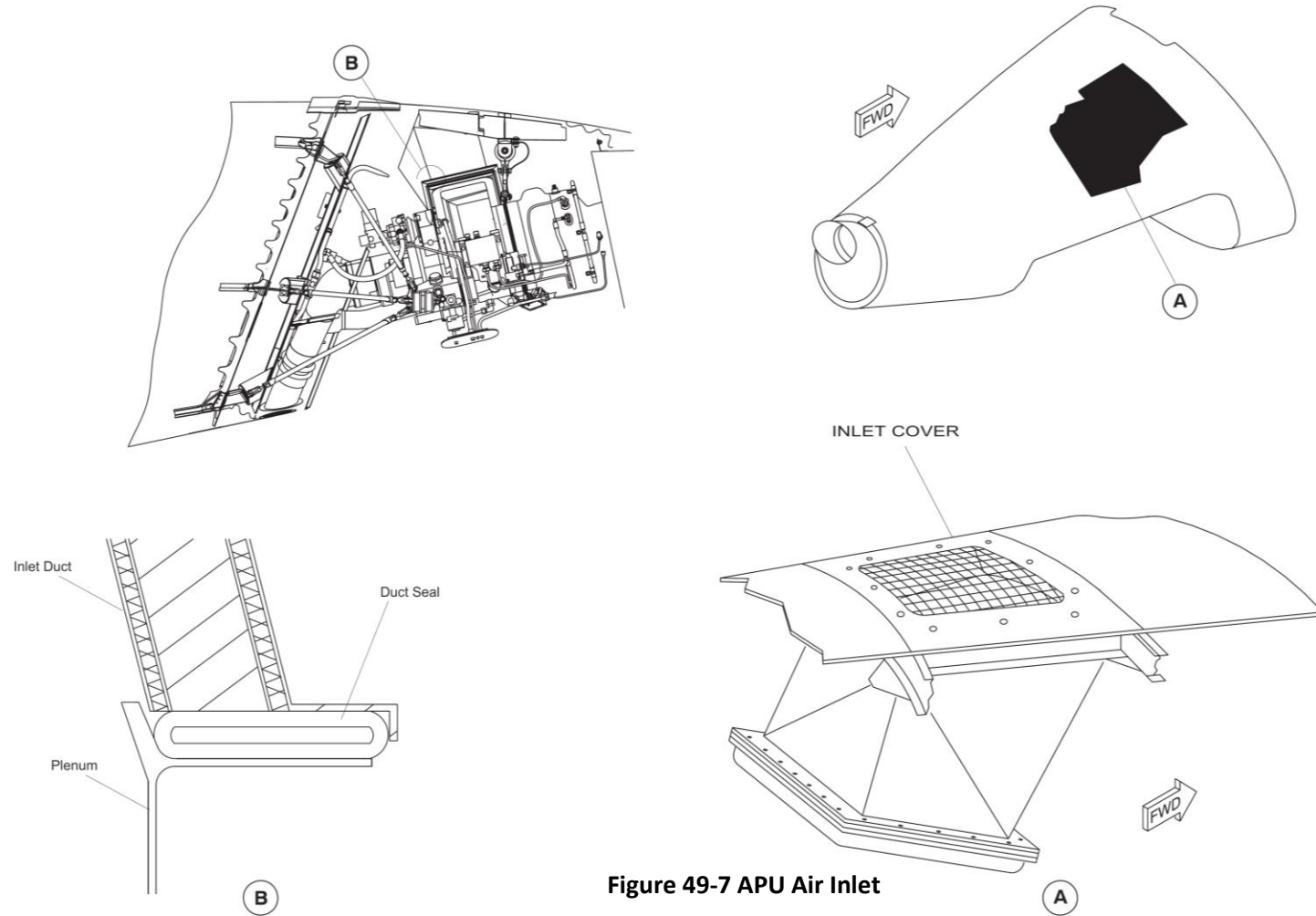


Figure 49-7 APU Air Inlet

APU Drains

[Refer Figure 49-8](#)

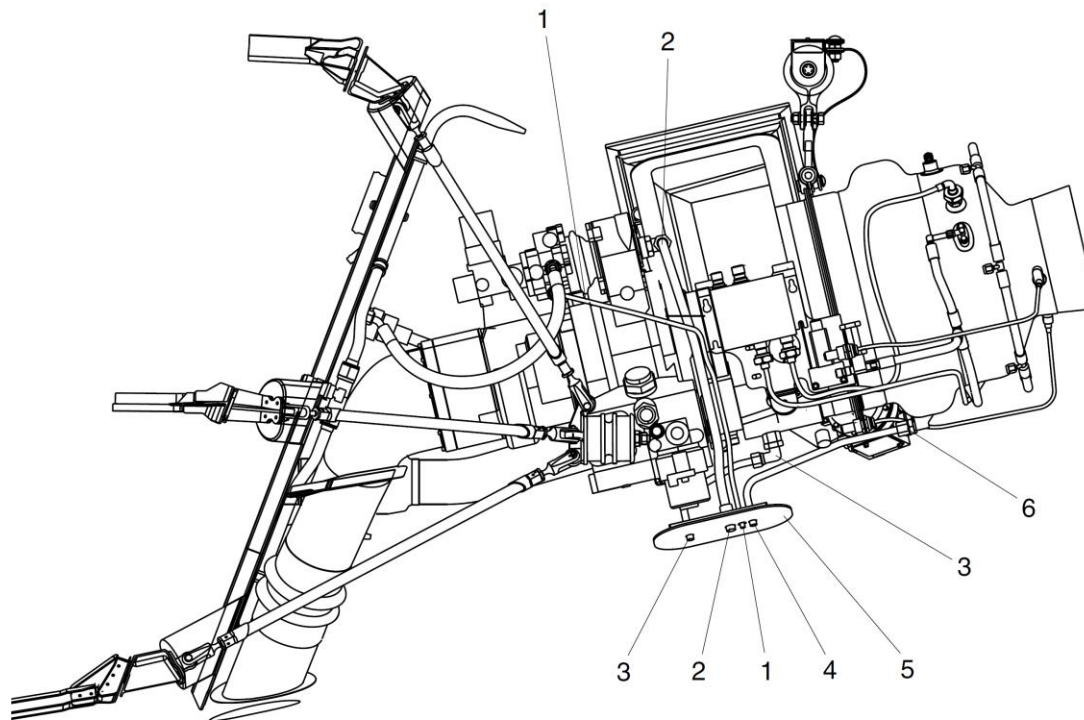
Drain connections are supplied for:

- Fuel pump drive
- Gearbox Breather
- Inlet plenum
- Combustor housing.

These connections have stainless steel tubes to connect them to the drain mast. The drain mast is titanium, aerofoil shaped and protrudes through the

left door of the tailcone. The mast has a separator plate to segregate the inlet plenum (front) drain from the other drains to prevent ingress of flammable fluids into the inlet. The combustor housing drain valve is threaded into the combustor drain boss. The valve is closed during engine operation by air pressure in the combustor housing. The valve is spring-loaded open when the engine is shut down, to let any accumulation of fuel in the combustor drain into the combustor drain system.

All intercostal and frames in the tailcone and doors have drain cutouts or holes at the lowest point to prevent the accumulation of liquids.



LEGEND

1. Fuel Pump Drain.
2. Gearbox Breather.
3. Inlet Muff Drain.
4. Combustor Drain.
5. Drain Mast Bracket.
6. Combustor Drain Valve.

Figure 49-8 APU Drains

Engine

Introduction

The engine assembly of the APU supplies shaft power to operate the Direct Current (DC) Generator, and bleed air to the Environmental Control System (ECS).

General Description

The Engine Assembly is a centrifugal compressor, gas turbine engine. A Full Authority Digital Electronic Controller (FADEC) controls operation of the Engine Assembly.

- Assembly, Engine

Detailed Description

[Refer Figure 49-9](#)

The engine assembly has three modules that follow:

- Turbine Assembly
- Combustor Assembly
- Gearbox Assembly

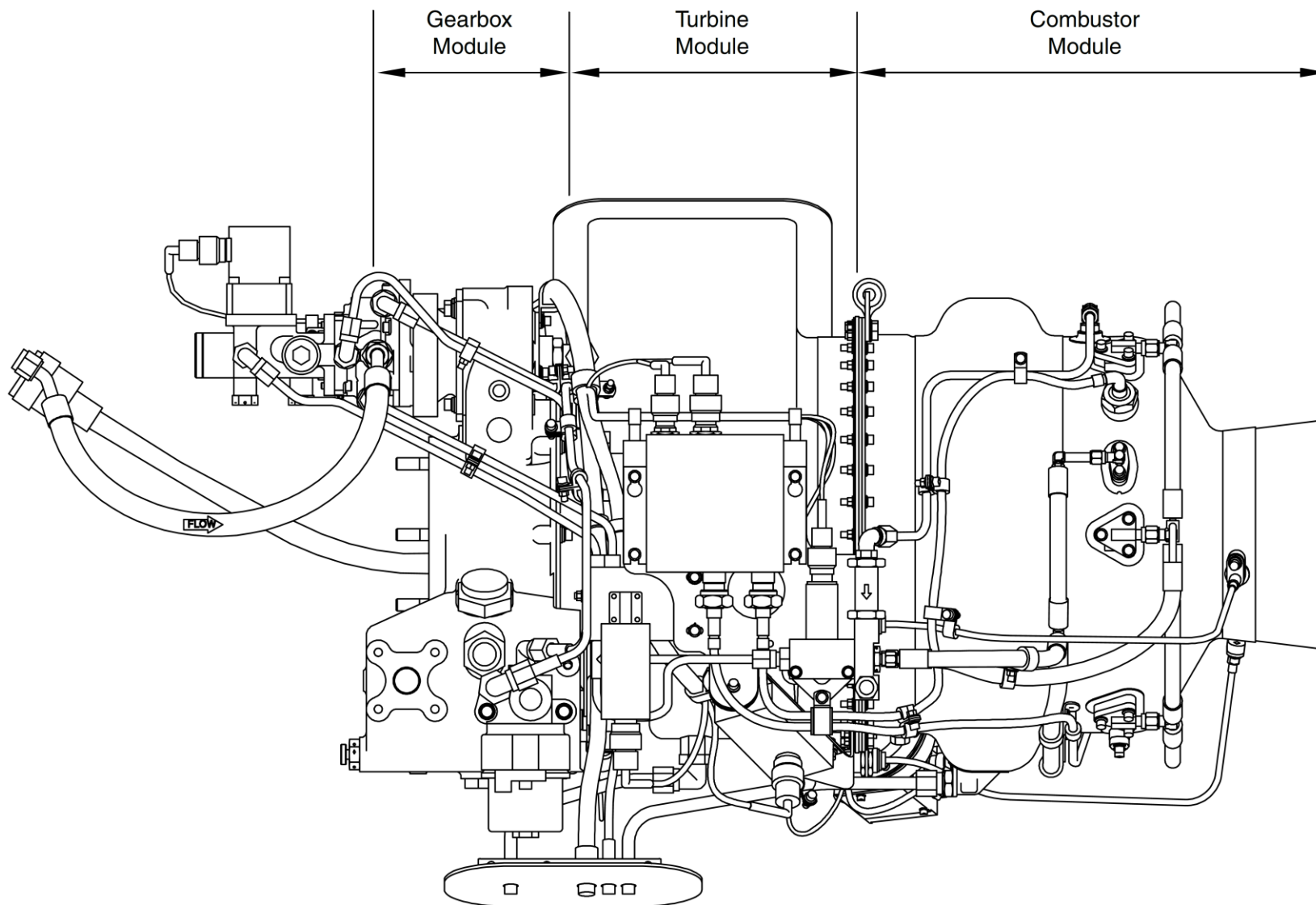


Figure 49-9 APU Engine Modules

Engine – Component Description

Turbine Assembly

The turbine assembly has the component that follows:

- Air Inlet Housing
- Rotor Assembly
- Turbine Nozzle/Containment Assembly

[Refer Figure 49-10](#)

Air Inlet Housing

The APU Air Inlet Housing assembly interfaces with the aircraft air inlet system to supply ambient air to the APU compressor. The Air Inlet Assembly is constructed of molded phenolic resin and is installed around the APU air inlet housing. A mesh screen is located inside the Air Inlet Assembly and is attached to the air inlet housing. The mesh screen is installed to prevent Foreign Object Damage (FOD).

The Air Inlet Housing supplies air to the compressor and gives a rigid support between the Gearbox and Combustor Assembly. The housing also has an internal attachment and support for the Rotor Assembly, Diffuser and Turbine Nozzle Containment Assembly. The housing also provides containment for the Compressor wheel of the Rotor Assembly.

Rotor Assembly

The single stage rotor is a balanced assembly that has an overhung radial Compressor Wheel and Turbine Wheel positioned on a two-bearing shaft system. The bearings are located in the air intake housing. A stationary segmented seal plate between the Compressor and Turbine Wheel prevents compressor discharge air from bypassing the Combustor.

Turbine Nozzle/Containment Assembly

The Nozzle/Containment assembly has a containment ring and a Turbine Nozzle. The Turbine Nozzle is attached to the containment ring by seven radial pins. Four bolts that pass through the diffuser and the air intake housing secure the complete assembly by fasteners on the forward side of the air

intake housing. Inside the turbine nozzle are 21 vanes, which direct the high velocity gas stream to the turbine wheel. Hot gas flows through the wheel and discharges axially through the turbine exhaust. Rotational energy derived from the hot gas flow stream by the turbine wheel drives the compressor and the shaft loads imposed by the generator and APU.

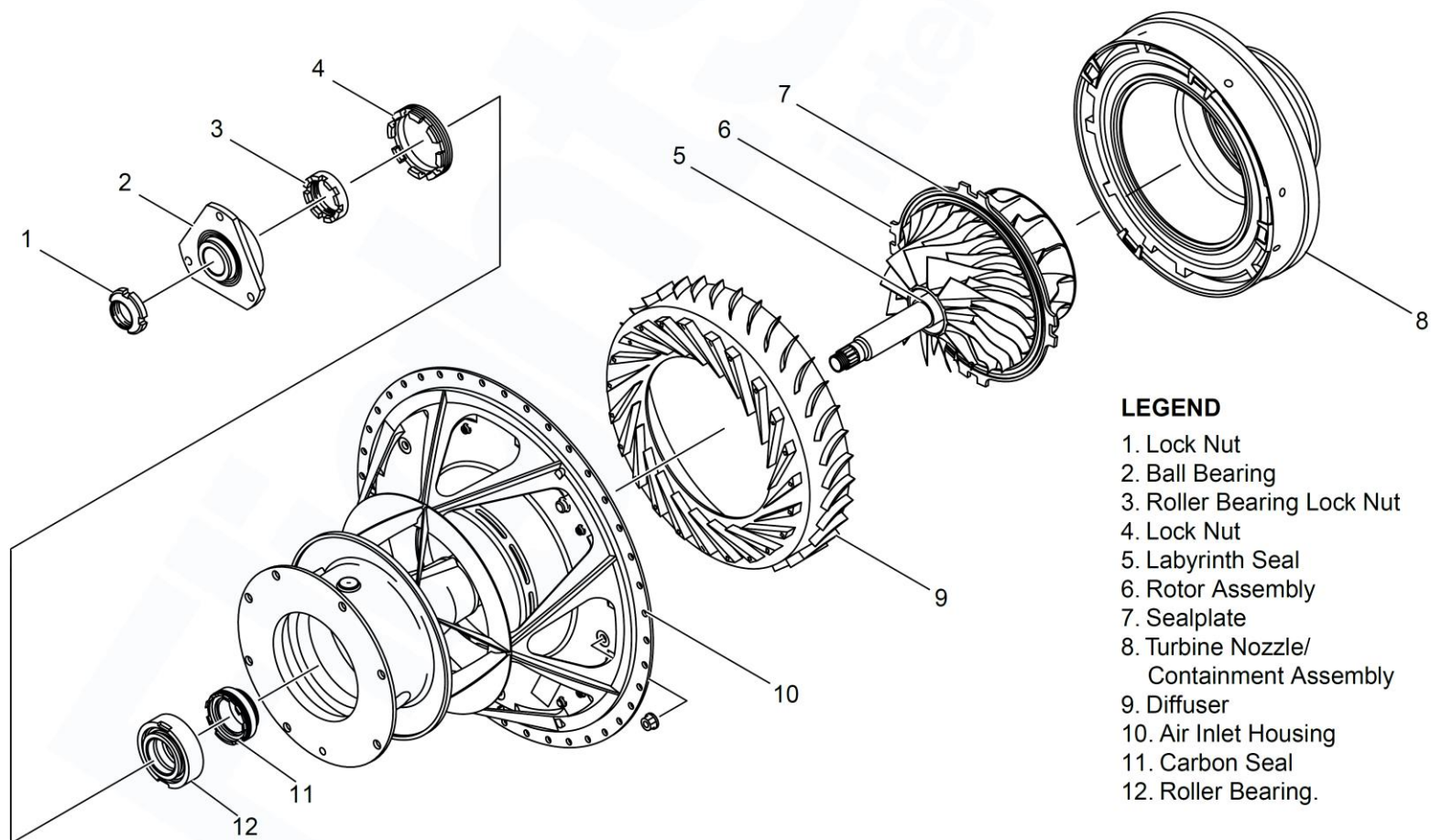


Figure 49-10 Combustor and Turbine Assemblies

Combustor Assembly

[Refer Figure 49-11](#)

The combustor assembly includes the components that follow:

- Combustor Housing
- Combustor Chamber

A bleed port is supplied on the combustor housing for the extraction of bleed air.

The combustor housing is attached to the air inlet housing with bolts. Compressor air is discharged into the combustor housing which forms a plenum surrounding the combustor chamber. Bleed air is ducted off this plenum when required, and the remainder of the air enters the combustor chamber through dilution holes. Six air atomizing fuel injectors directed circumferentially into the combustor chamber provide for complete combustion, thorough mixing, and uniform temperature distribution of the hot gas entering the turbine nozzle. The combustor housing itself is a one-piece flow-formed outer housing.

The combustor drain valve is threaded into the combustor drain boss. The valve is closed during engine operation by air pressure in the combustor housing. The valve is spring loaded open when the engine is shut down, to let any accumulation of fuel in the combustor drain into the combustor drain system.

The combustor chamber is an annular combustor and is located inside the combustor housing. Two start nozzles, six air atomizing fuel injectors and two igniters go through the combustor chamber. Combustion takes place inside the chamber near the fuel injectors. Dilution air is added downstream where a thorough mixing of fuel and air occurs to produce a uniform hot gas stream. The chamber walls are film cooled by air introduced from behind multiple cooling strips that distributes the air over the inner surface of the chamber.

LEGEND

1. Combustor Liner
2. Ignitor Boss
3. Start Fuel Nozzle Boss
4. Main Fuel Nozzle Boss
5. Pin Liner Boss
6. Combustor Housing
7. Combustor Drain Boss
8. Bleed Air Port
9. Start Fuel Nozzle (2)
10. Main Fuel Injector (6)
11. Ignitor (2)
12. Combustor Drain Valve
13. O-Ring.

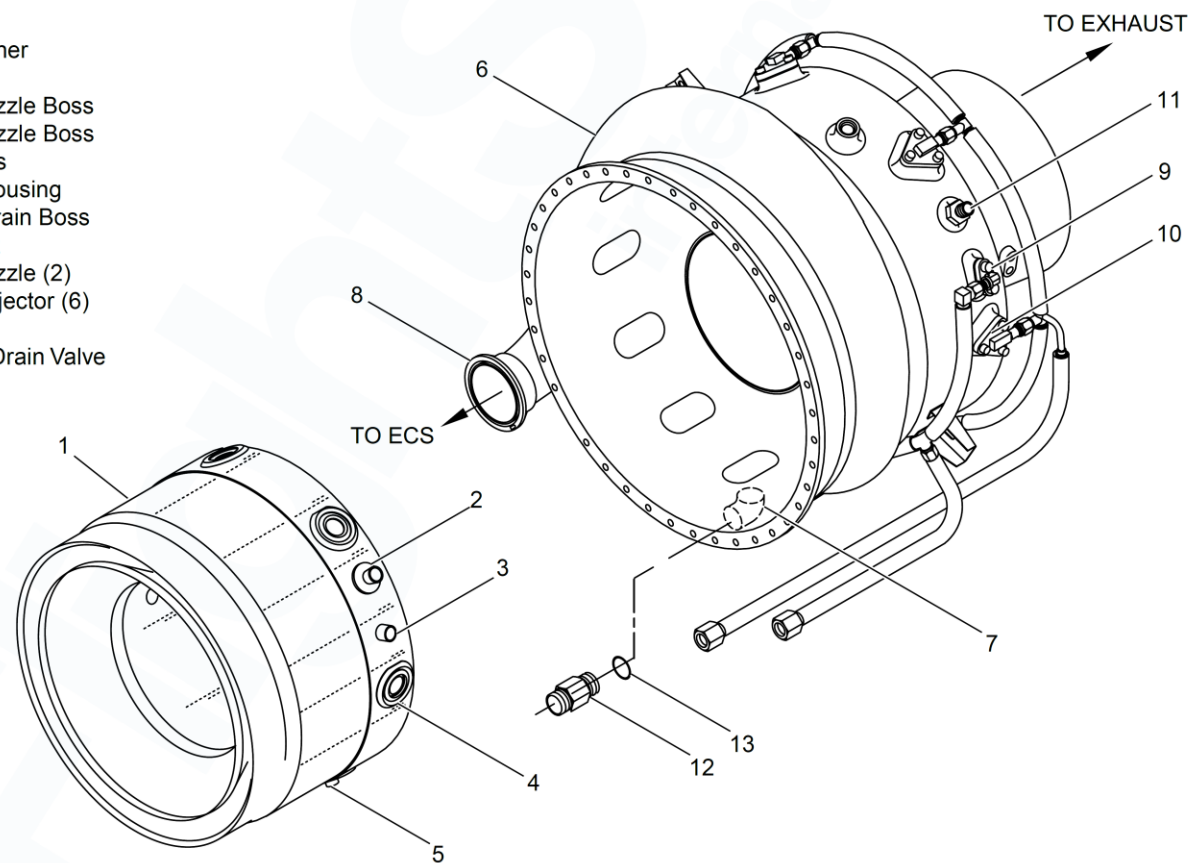


Figure 49-11 Combustor Assembly

Gearbox Assembly

[Refer Figure 49-12](#)

The gearbox housing is a machined aluminum casting with a fireproof oil sump. All gear shafts are supported in the housing by roller and ball bearings. The aft section of the gearbox provides a flange for the attachment to the engine turbine module. The gearbox has a main output pad for the starter/generator and a second pad for the fuel control/pump assembly. Bosses on the housing allow for the mounting of high oil temperature and low

oil pressure switches.

The gearbox module provides the reduction of the turbine rotor speed to speeds required for starter/generator, fuel pump, and oil pump drives. The input quill drives a set of fixed star gears, which transfer power to a ring gear. The oil pump gear is driven from the output shaft. The fuel pump gear is driven through a single idler gear. The fuel pump gear includes an air/oil separator for gearbox venting.

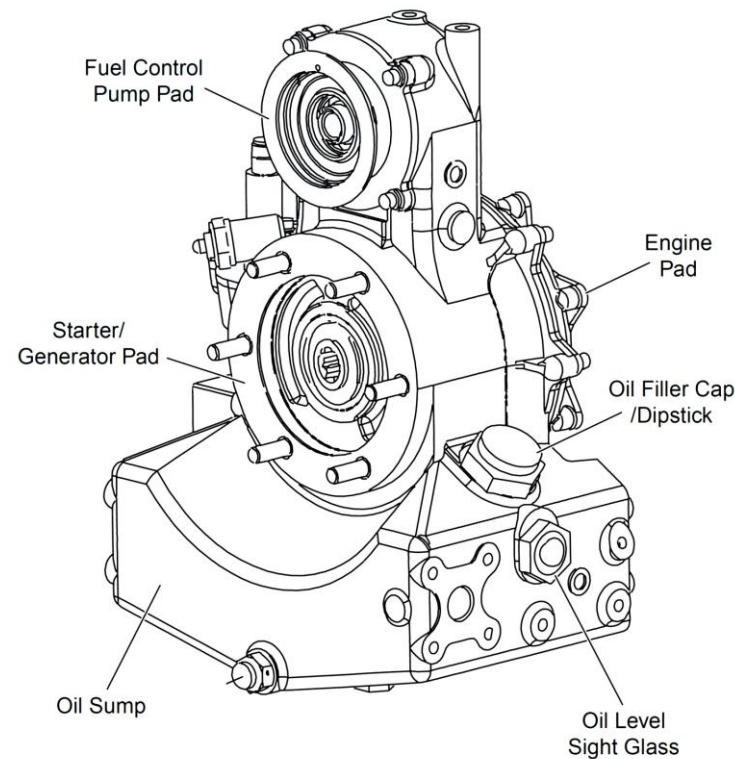


Figure 49-12 Gearbox Assembly

Engine Fuel and Control

Introduction

The Engine Fuel and Control system monitors and supplies the fuel to the Auxiliary Power Unit (APU). The system automatically compensates for altitude and APU operations.

General Description

Fuel delivery to the APU electronic fuel control and pump assembly (Fuel Control) is supplied by the aircraft fuel system. The Electronic Control Unit (ECU) supplies the signals that operate the fuel control.

The system has the components that follow:

- Assembly, Fuel Control
- Valve, Fuel-Start
- Valve, Fuel-Main
- Check Valve, Combustor Drain
- Divider, Flow
- Check Valve, Purge
- Nozzle, Fuel-Start
- Nozzle, Fuel-Main
- Manifold, Fuel-Start
- Manifold, Fuel-Main
- Assembly, Fuel Filter
- Element, Fuel Filter

Detailed Description

[Refer Figures 49-13 & 49-14](#)

The gearbox driven fuel control receives fuel from the aircraft fuel system. Fuel flows through the low pressure centrifugal fuel pump to the fuel/oil cooler, fuel filter, high pressure fuel pump, the fuel servo valve and to the fuel solenoid manifold assembly. Excess fuel is circulated through the relief valve across the high pressure fuel pump. A differential pressure valve is in parallel with the fuel servo valve to maintain a constant differential pressure across the fuel servo valve. This results in a fuel flow linear relationship to the servo

current command from the Full Authority Digital Electronic Control (FADEC).

The FADEC controlled fuel flow provides efficient starting and operation of the APU.

The start sequence begins with the servo valve energized open at start up.

The start fuel valve is energized open at 3% gas generator (Ng) speed and fuel flows to the two start nozzles and is ignited by the igniters. When 25% Ng speed is reached the main fuel valve is energized open and fuel flows to six additional main fuel nozzles. A flow divider valve that is located in the fuel solenoid manifold assembly opens when fuel pressure reaches 300 psid (2068.4 kPad). The valve makes sure proper fuel pressure is maintained to the start nozzles.

The start fuel valve stays open to 70% Ng speed and then is de-energized closed. At this time, the start fuel check valve, located downstream of the start fuel valve, opens to remove residual fuel from the Start Fuel manifold and nozzles. The start fuel check valve is opened by APU compressor air pressure (PCD) when fuel pressure is no longer present. The valve gives a continuous PCD air flow through the nozzles during APU operation.

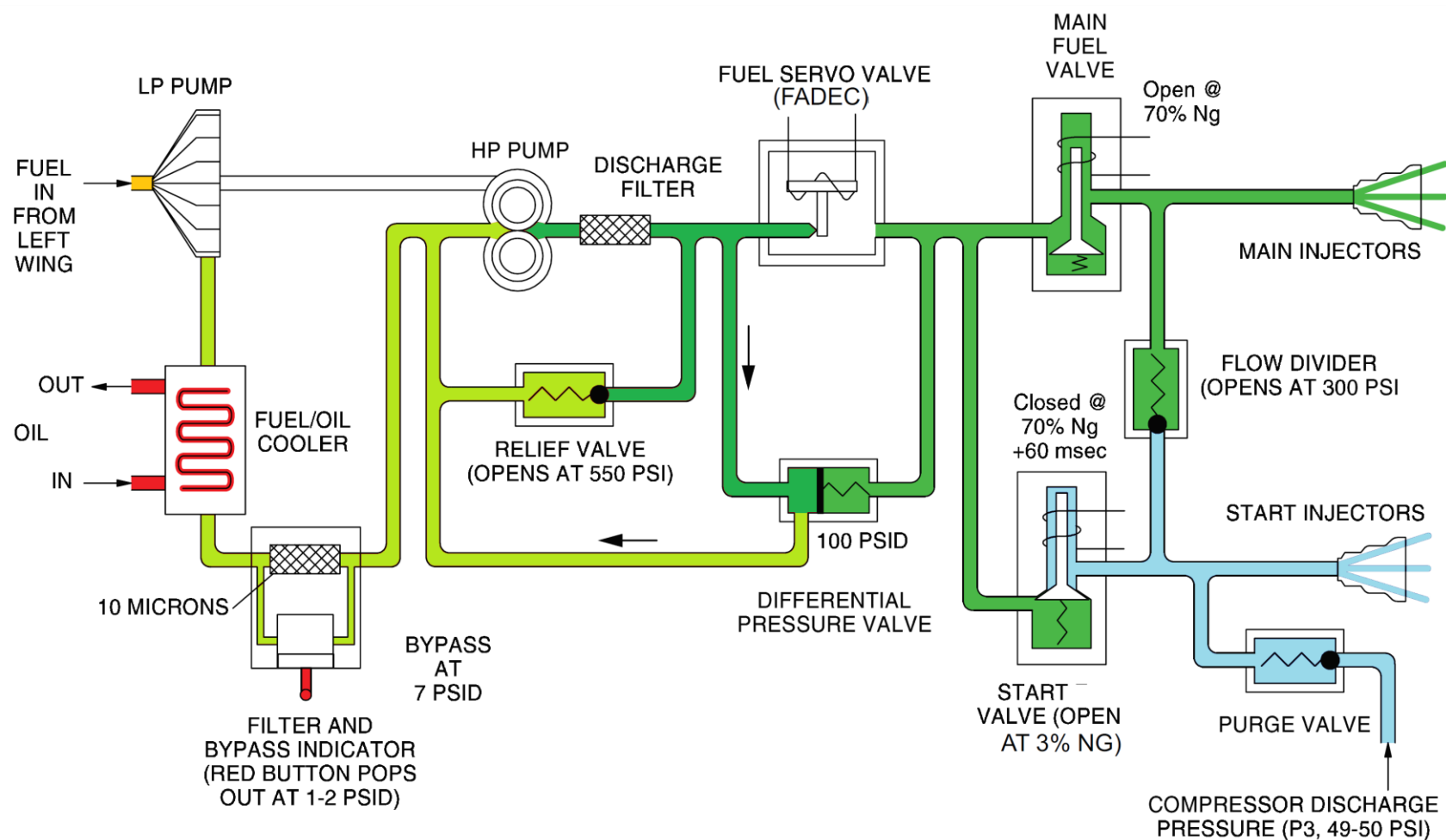
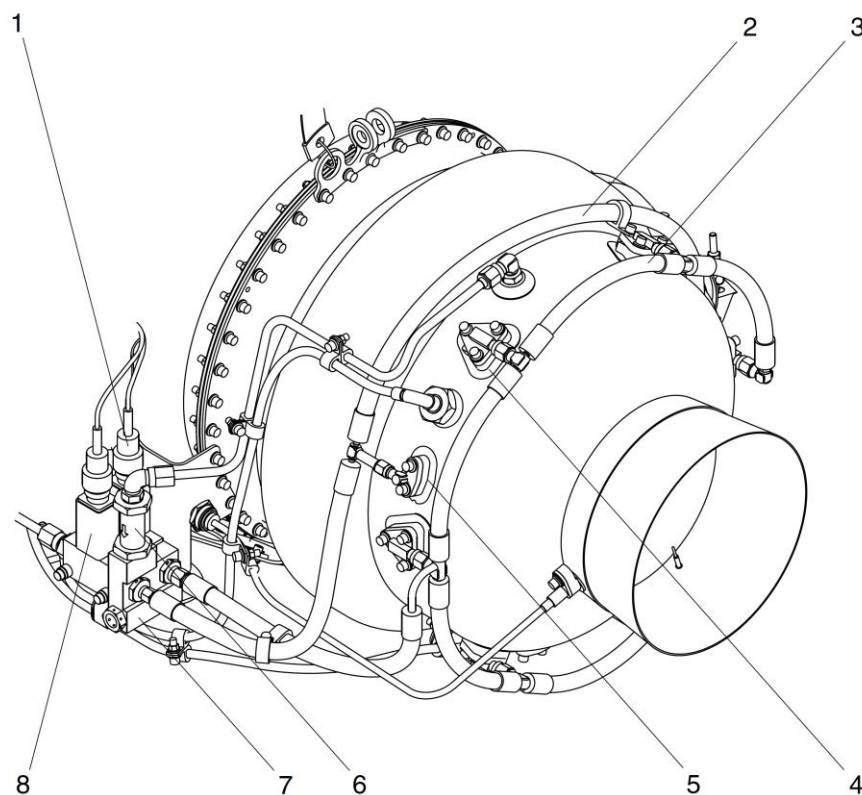


Figure 49-13 Fuel System Schematic



LEGEND

1. Main Fuel Solenoid Valve.
2. Start Fuel Manifold.
3. Main Fuel Manifold.
4. Main Fuel Nozzle.
5. Start Fuel Nozzle.
6. Start Fuel Check Valve.
7. Fuel Solenoid Outlet
Manifold Block Assembly.
8. Start Fuel Solenoid Valve.

Figure 49-14 APU Fuel System Detail

Engine Fuel and Control – Component Description

Assembly, Fuel Control

[Refer Figure 49-15](#)

The electronic fuel control assembly is located on the fuel pump. The assembly contains a fuel servo valve and a differential pressure valve.

The servo valve is energized open at start up. The fuel servo receives milliamp

signals from the FADEC to maintain 100% Ng speed during all APU load conditions. In the shutdown sequence FADEC maintains current to the fuel servo valve for 0.5 seconds and then de-energizes the valve.

The differential pressure valve is in parallel with the fuel servo valve to maintain a constant differential pressure across the fuel servo valve. This results in a fuel flow linear relationship to the servo current command from the FADEC.

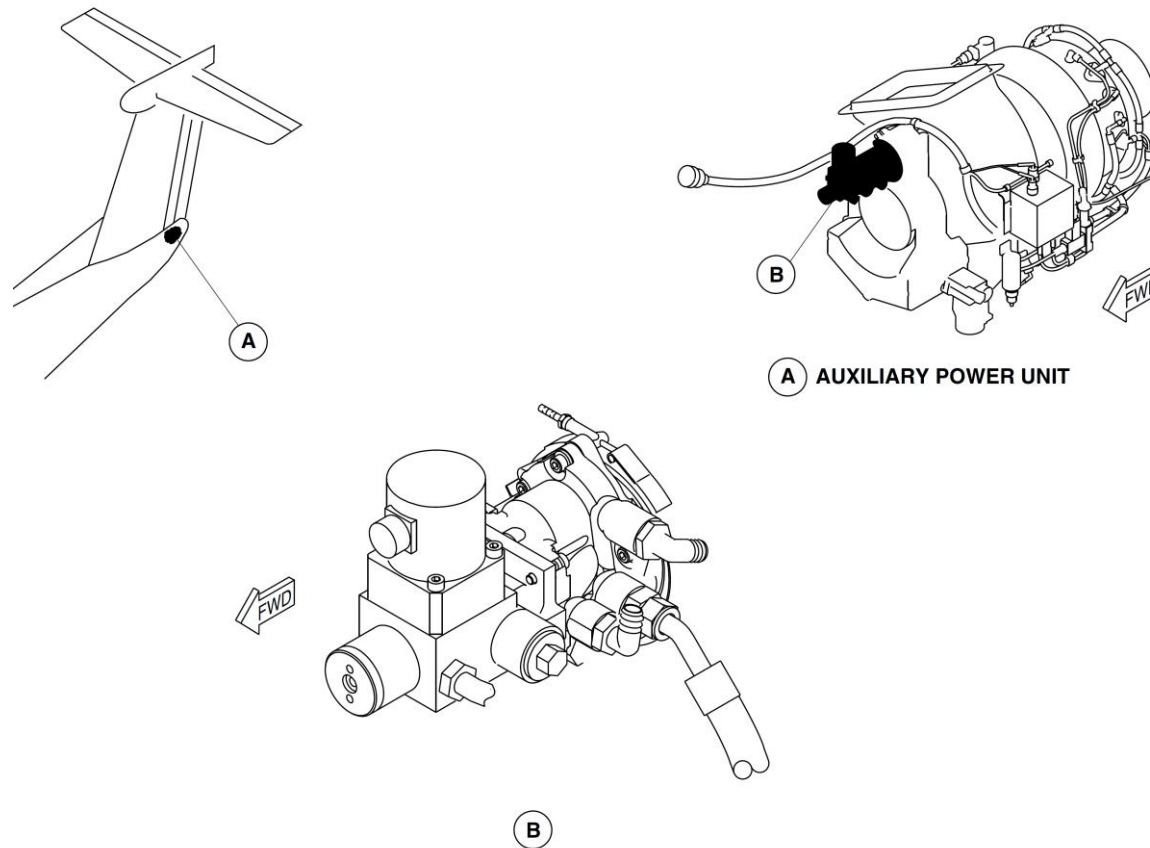


Figure 49-15 Fuel Control Assembly

Valve, Fuel-Start

[Refer Figure 49-16](#)

The start fuel valve is installed on the fuel solenoid manifold assembly. The valve is energized open at 3% Ng speed and fuel flows to the two start nozzles and is ignited by the igniters. The valve stays open to 70% Ng speed and then is de-energized closed.

Valve, Fuel-Main

[Refer Figure 49-16](#)

The main fuel valve is installed on the fuel solenoid manifold assembly. The valve is energized open when 25% Ng speed is reached and the fuel flows to the six main fuel nozzles. The valve stays open during engine acceleration and throughout engine operation.

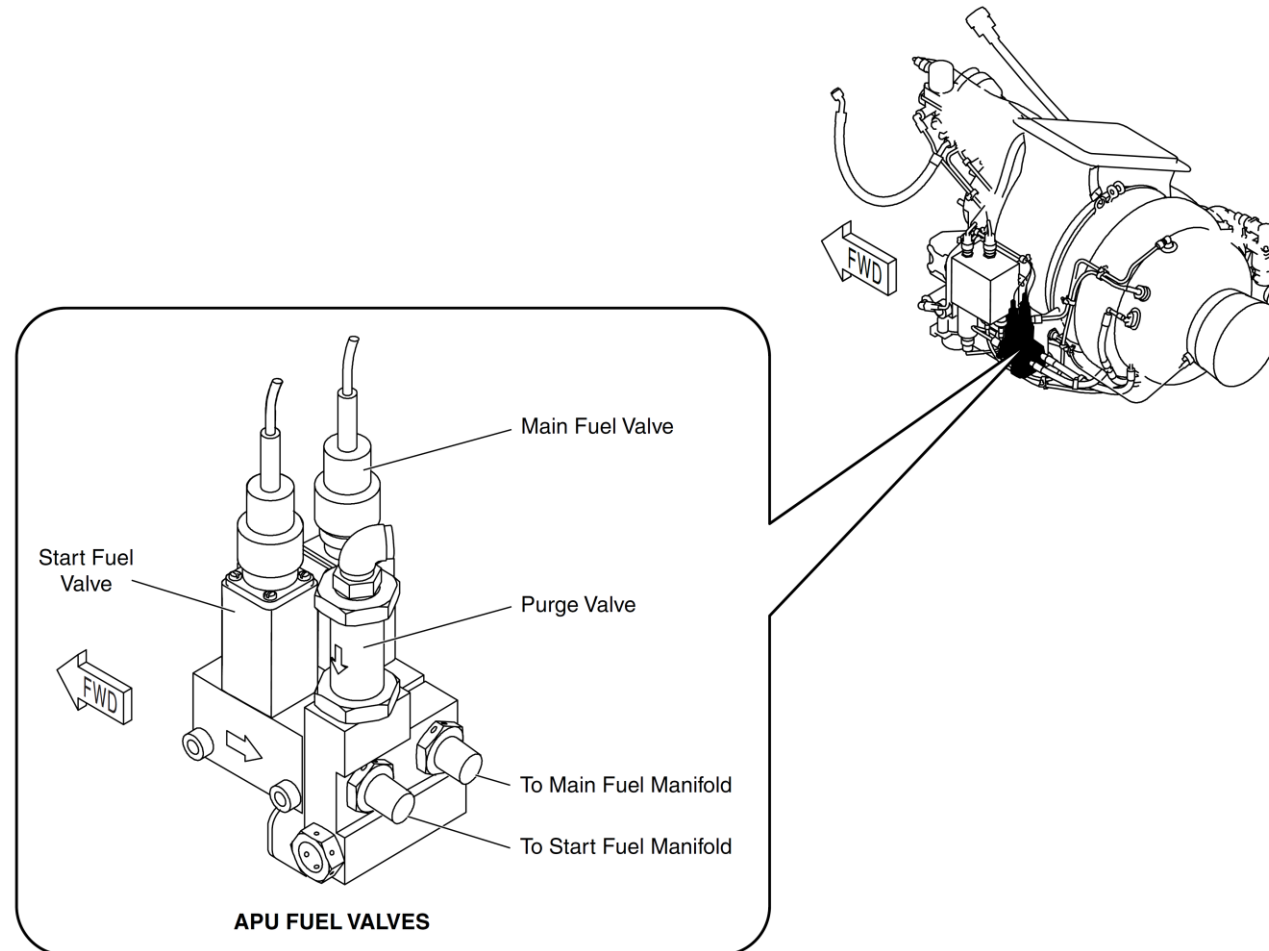


Figure 49-16 APU Fuel Valves

Check Valve, Combustor Drain

[Refer Figure 49-17](#)

The valve is threaded into the combustor drain boss and is closed, during engine operation, by air pressure in the combustor housing. The valve is spring loaded open when the engine is shut down, to let any accumulation of fuel in the combustor drain into the combustor drain system.

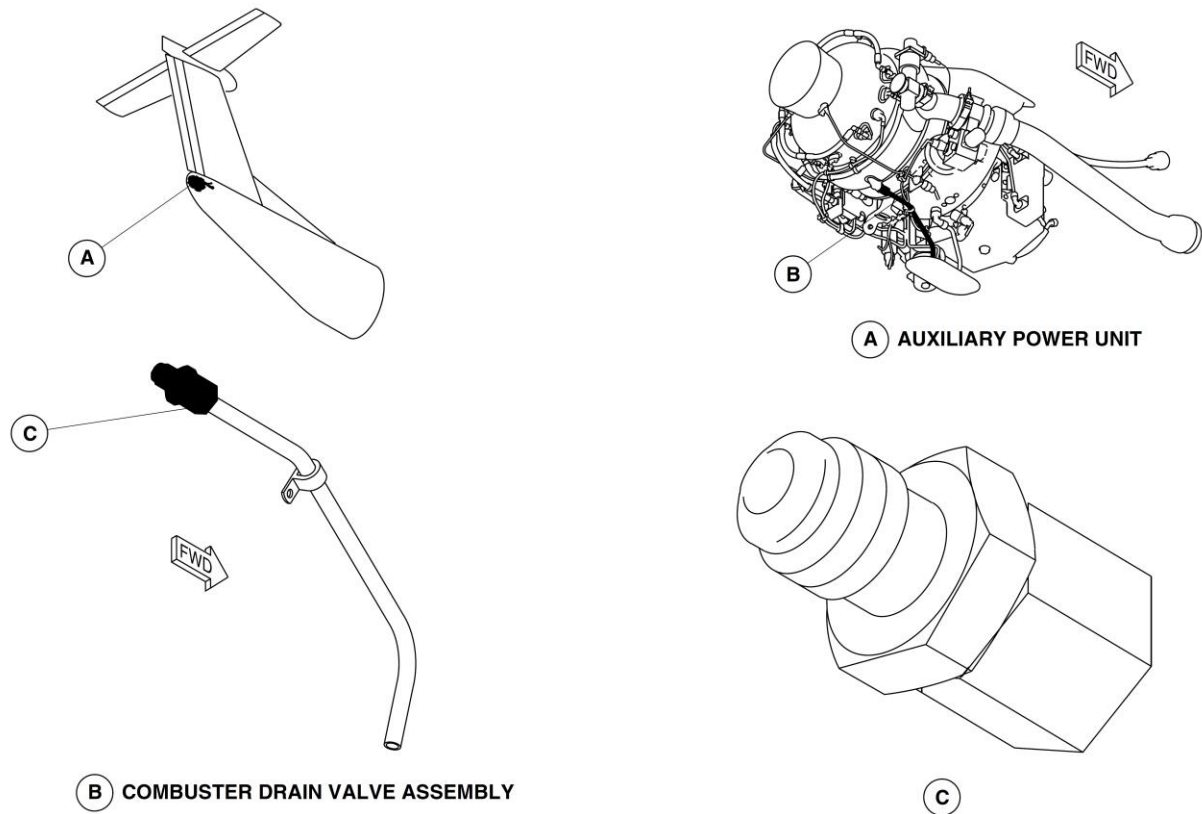


Figure 49-17 Combustor Drain Check Valve

Divider, Flow

[Refer Figure 49-18](#)

The flow divider valve is located in the fuel solenoid manifold assembly. The valve opens when fuel pressure reaches 300 psid (2068.4 kPad) to make sure proper fuel pressure is maintained to the start nozzles.

Check Valve, Purge

[Refer Figure 49-18](#)

The valve is located downstream of the start fuel valve and opens to purge residual fuel from the start fuel manifold and nozzles. It is opened by APU compressor air pressure when fuel pressure is no longer present. The valve supplies a continuous air flow through the nozzles during APU operation.

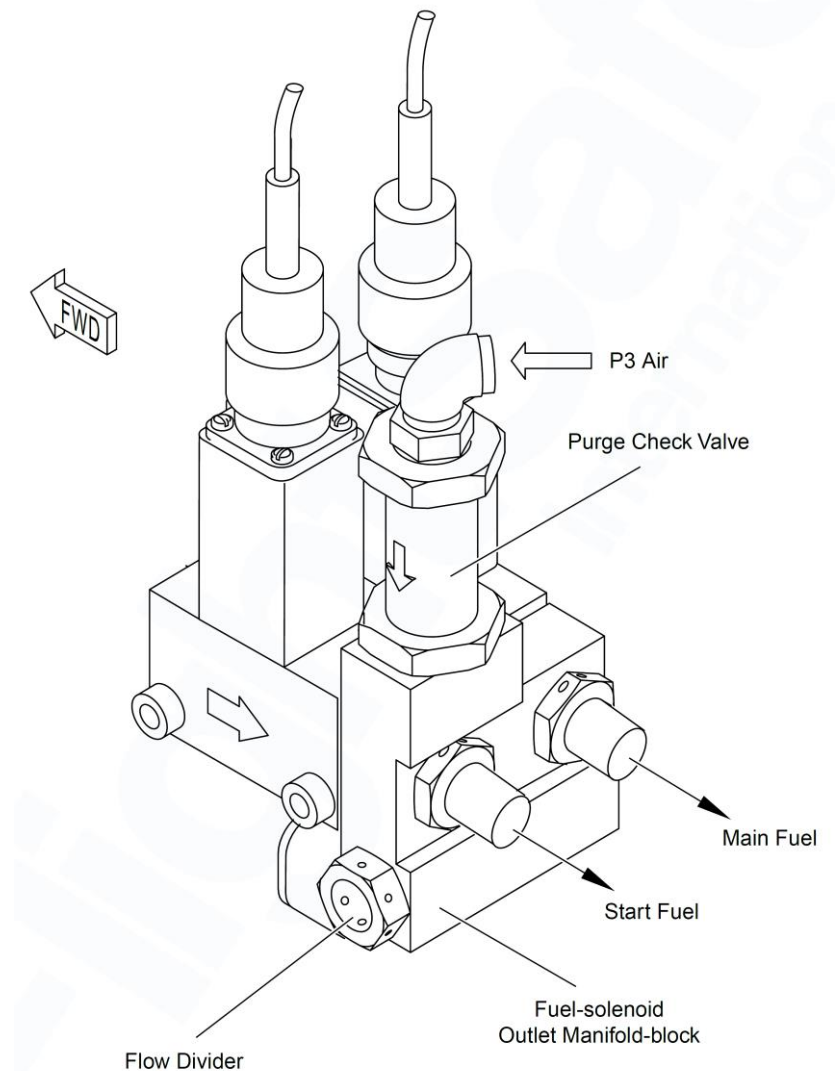


Figure 49-18 Flow Divider & Purge Check Valve

Refer Figure 49-19

Nozzle, Fuel-Start

The two start fuel nozzles are threaded into individual nozzle assemblies and installed in the combustor housing with bolts.

Nozzle, Fuel-Main

The six fuel nozzle assemblies are installed in the combustor housing with bolts.

Manifold, Fuel-Start

The flexible start fuel manifold is located on the engine combustor housing and supplies fuel to the two start fuel nozzles.

Manifold, Fuel-Main

The main fuel manifold is flexible and is located around the engine combustor housing. The manifold supplies fuel to the six fuel nozzle assemblies.

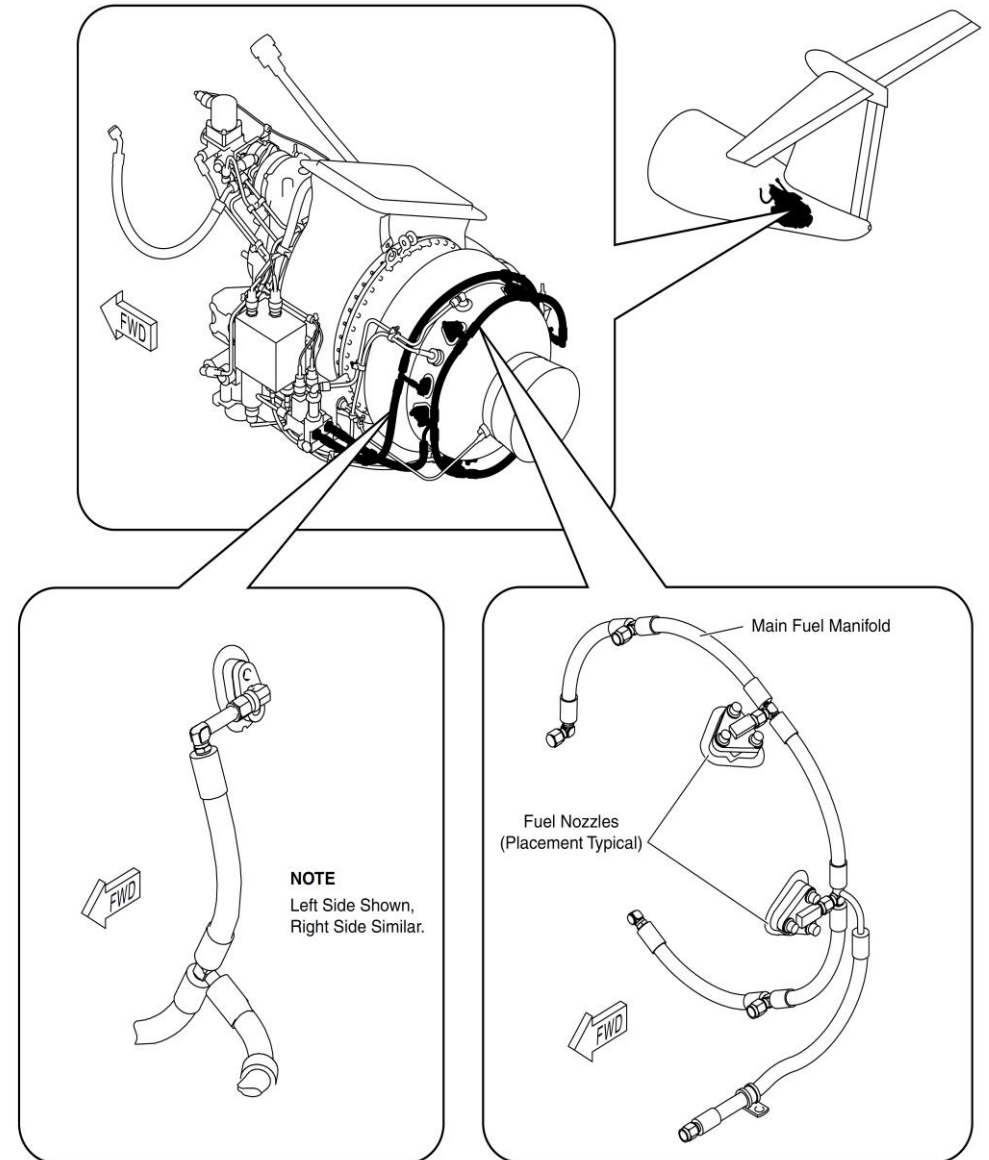


Figure 49-19 Start and Main Fuel Manifolds and Nozzles Locator

Assembly, Fuel Filter

[Refer Figure 49-21.](#)

The low-pressure fuel filter protects the fuel control from contamination. The filter differential pressure indicator and bypass valve are located on the side of the filter housing. The impending bypass indicator gives a visual warning when fuel flow through the filter becomes restricted and a pressure differential of 1 to 2 psid (6.89 to 13.8 kPad) occurs. The filter bypass valve will open at a pressure differential of 7 psid (48.3 kPad) to let fuel bypass the filter and maintain flow through the system.

Element, Fuel Filter

[Refer Figure 49-20](#)

The filter is a full flow type with a replaceable 10-micron filter element.

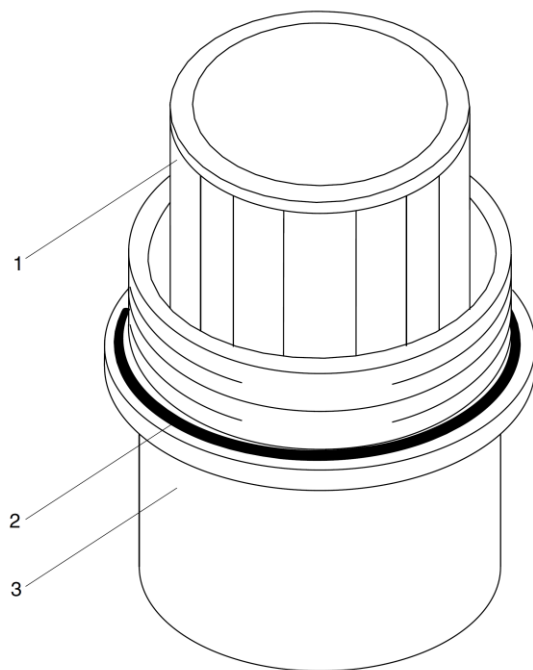
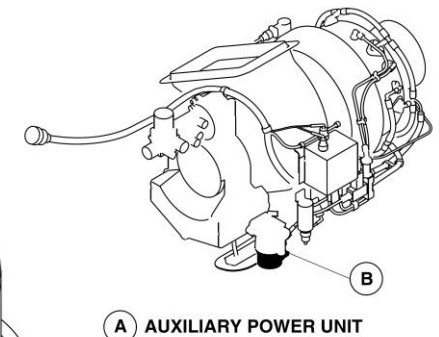
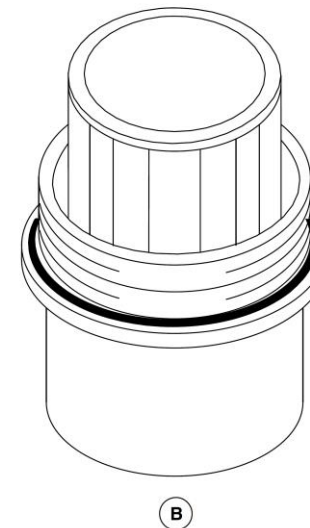
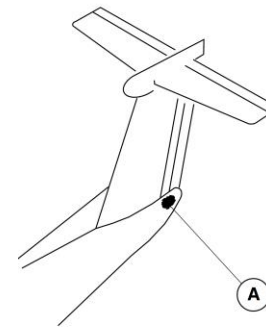


Figure 49-20 Filter Element

LEGEND

1. Fuel Filter Element.
2. O-Ring.
3. Filter Bowl.



A AUXILIARY POWER UNIT

Figure 49-21 Fuel Filter Assembly

Engine Controls

Introduction

The Auxiliary Power Unit engine control system controls the -operation of the APU. The system also records all operations of and -events associated with the APU.

General Description

The engine controls include the components that follow:

- APU Control Panel
- Full Authority Digital Electronic Control (FADEC)

Engine Controls - Component Description

APU Control Panel

[Refer Figure 49-22](#)

The APU Control Panel is located in the flight compartment and -installed in the center overhead panel.

PWR SWITCHLIGHT (alternate action)

- PUSH – arms APU start circuits and opens the APU fuel valve, APU FUEL VALVE OPEN light (green) shown on the APU fire protection panel – only arms aircraft on ground, no fire detected, and EXTG switch not selected
- RUN segment (green) – APU is at operating speed after the START switchlight is pushed
- PUSH – RUN segment (blank) – closes APU fuel valve, APU FUEL VALVE CLOSED light (white) shown on the APU fire protection panel, to stop APU
- FAIL – a failure is detected and the APU automatically stops.

START SWITCHLIGHT (alternate action)

- PUSH – STARTER segment (amber) – starts automatic APU start sequence
- STARTER segment (blank) – APU start sequence complete or stopped

GEN SWITCHLIGHT (alternate action)

- PUSH – ON segment (green) – APU starter-generator is supplying DC

power

- PUSH – ON segment (blank) – selects APU generator off line
- WARN segment (amber) – APU starter-generator off line.

BL AIR SWITCHLIGHT (alternate action)

- PUSH – OPEN segment (green) – APU bleed air valve open– APU bleed air supplies air to the ECS if engine bleed air is off
- PUSH – OPEN segment (blank) – APU bleed air valve is closed – selects APU bleed air off

GEN OHT ADVISORY LIGHT

- GEN OHT segment (amber) – APU starter-generator overheat condition – APU automatically shuts down

When the Power (PWR) switch is selected, 28 Vdc electrical power is supplied to the FADEC, enabling APU functions and the control panel advisory lights.

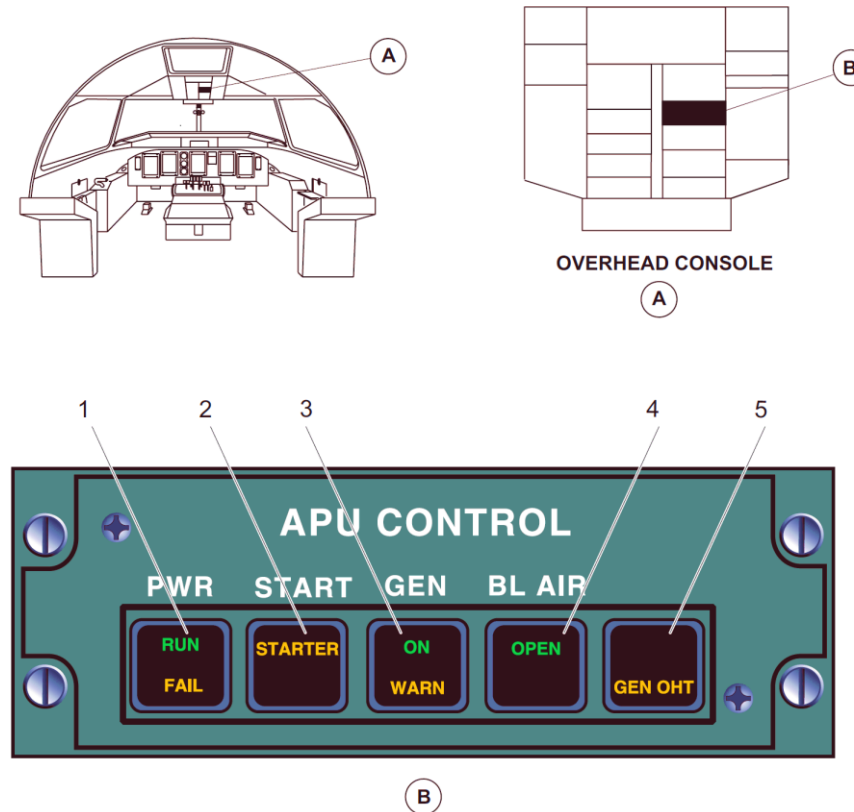
The START button is pushed to begin the auto start sequence. The FADEC commands the generator control unit to begin the start sequence.

After the APU is started and the RUN advisory is displayed on the PWR button, the generator can be set to ON by the generator button and bleed air can be selected, by pressing OPEN on the BL AIR button.

The FAIL light on the PWR button shows that a malfunction or failure has been sensed in the APU system.

The WARN light on the GEN button shows that the APU generator is not on line.

The GEN OHT light on the APU control panel comes on to warn of a generator overheat. The APU caution light in the flight compartment will also come on in this situation.



LEGEND

1. Power Run\Failure Pushbutton Annunciator Switch.
2. Starter Pushbutton Annunciator Switch.
3. Generator On\Warning Pushbutton Annunciator Switch.
4. Bleed Air Open Pushbutton Annunciator Switch.
5. Generator Overheat Annunciator Light.

Figure 49-22 APU Control Panel

Full Authority Digital Electronic Control (FADEC)

[Refer Figure 49-23](#)

The FADEC is a microprocessor based controller, that can do control sequencing, condition monitoring, fault detection, fault isolation and protection of the APU system. The APU FADEC monitors the APU speed, oil temperature, oil pressure, and exhaust gas temperature. It controls the APU start up, fuel scheduling, manual and auto shutdown and overspeed protection. The FADEC transmits fault and maintenance data to the aircraft CDS through an ARINC 429 data link.

APU FADEC is located at the aft of the aircraft near the bottom center of the middle spar of the vertical stabilizer in the ECS compartment.

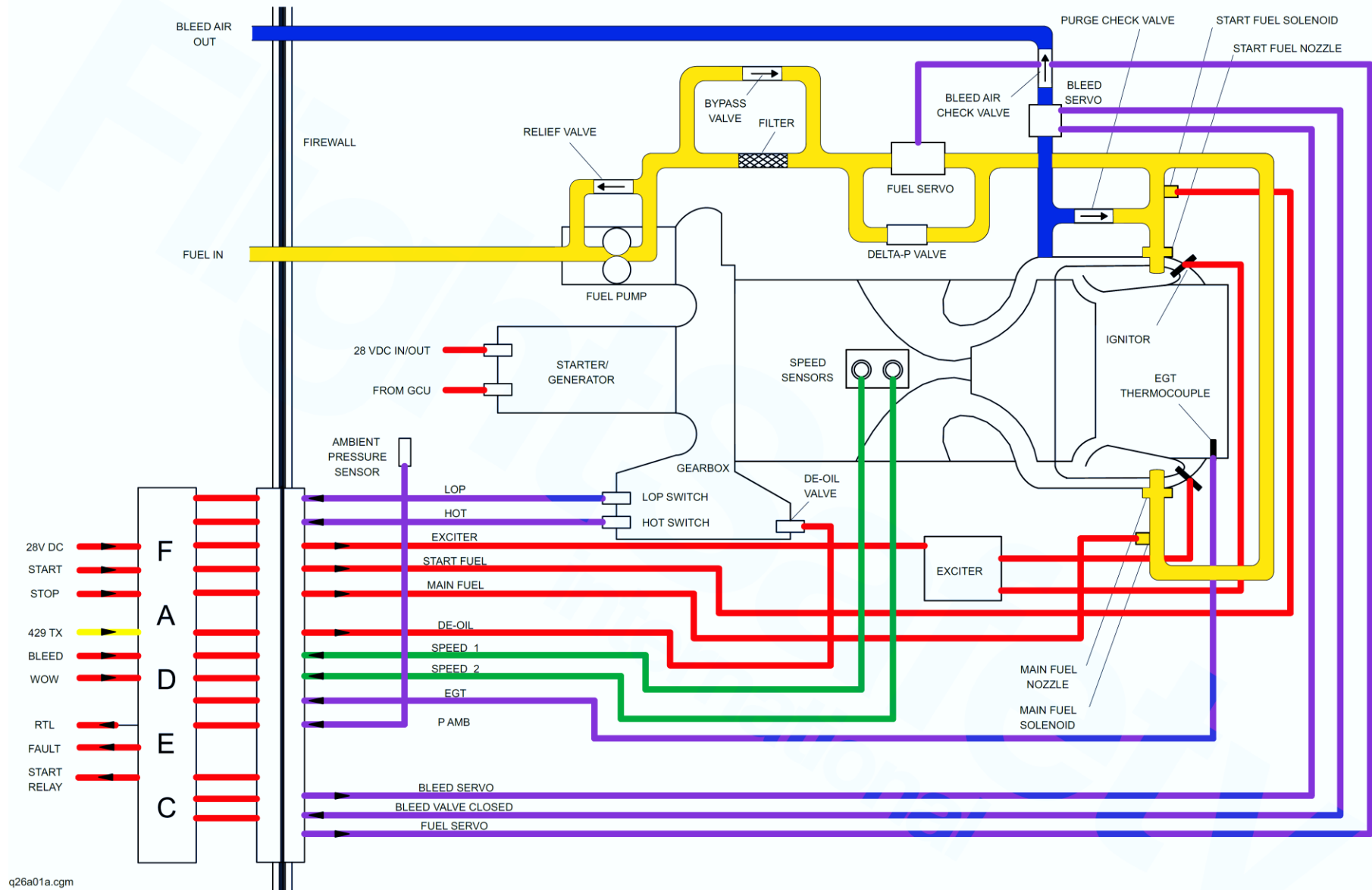


Figure 49-23 APU Control Schematic

Ignition and Starting

Introduction

The starting and ignition system includes the controls, indicators and switches for automatic starting, acceleration, operation and monitoring of the APU.

General Description

The system includes the components that follow:

- APU Control Panel
- Starter/generator
- Ignition Exciter
- Igniter Plugs

Detailed Description

The APU is started by selecting the START switch on the APU control panel. The start sequence and operating sequence are then automatically controlled by the APU Full Authority Digital Electronic Control (FADEC).

Ignition and Starting – Component Description

APU Control Panel

[Refer Figure 49-24](#)

The APU control panel is installed on the right hand side of the flight compartment overhead panel. Four switches and an annunciator are installed next to each other in one row. The panel is supplied with 5 Vdc background lighting when selected. The START switch is a momentary switch, the other switches are alternate actuation switches.

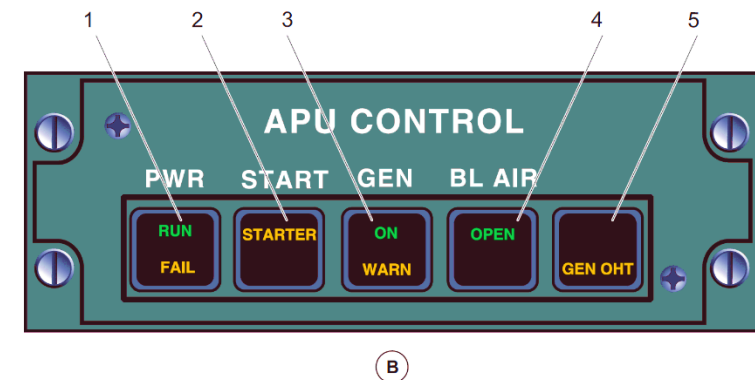
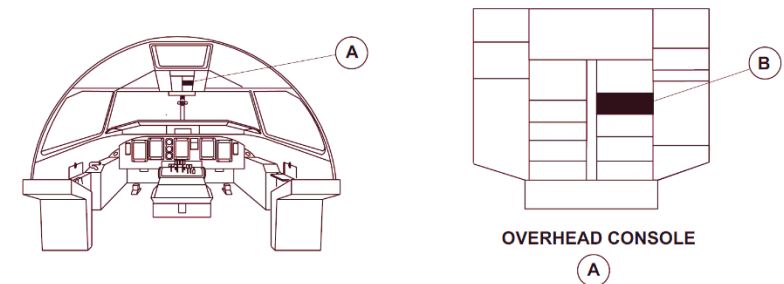
Selecting the power (PWR) switch applies 28 Vdc power to the FADEC, enabling APU functions and the control panel advisory lights. Pressing the START switch starts the auto start sequence. FADEC signals the Generator Control Unit to begin the start sequence.

After the APU has started and the RUN advisory light on the PWR switch has come on, the GEN ON switch selects the generator and BL AIR selects bleed air on.

Pushing the PWR switch, when the APU is operating, will close the fuel valves and shut down the APU. This is a normal shut down and will not turn on the advisory or caution lights.

GEN OHT turns on when the generator is hot.

FAIL turns on for any detected failure of the APU or APU control functions.



LEGEND

1. Power Run\Failure Pushbutton Annunciator Switch.
2. Starter Pushbutton Annunciator Switch.
3. Generator On\Warning Pushbutton Annunciator Switch.
4. Bleed Air Open Pushbutton Annunciator Switch.
5. Generator Overheat Annunciator Light.

Figure 49-24 APU Control Panel

Starter Generator

[Refer Figure 49-25](#)

The Starter/Generator is an electromagnetic rotating machine that has two modes of operation.

Starter Mode

The unit converts 28 VDC power into mechanical torque that is used to drive the APU rotor for engine starting.

Generator Mode

The unit converts mechanical power from the gearbox into 28 VDC. The modes of operation are controlled by the FADEC and the Generator Control Unit (GCU).

The 28 VDC, 400 A starter/generator is mounted on the forward face of the gearbox. The unit is mounted on a Quick Attach/Detach Adapter (QAD). The QAD incorporates the cooling air outlet. QAD is attached on six studs with self-locking nuts. A V-band clamp attaches the starter/generator to the QAD.

The drive spline is lubricated by air oil mist from the gearbox. Any oil leakage picked up by the cooling airflow to blown overboard through the discharge duct. The starter/generator is an air cooled unit with an integral cooling fan. The inlet is titanium with a metal mesh screen. Each end of the composite duct is secured and sealed with a fireproof silicone composite boot and hose clamps. The outlet duct is bolted to the QAD. The outlet duct has a fireproof silicon seal at the point where it contacts the right access door. A bimetallic strip in the stator assembly provides a means for indicating a starter/generator overheat.

At 50% N_g the starter is de-energized. When the APU reaches 95% speed plus 3 seconds the generator is available. The generator has the same generating capacity as the engine generators. The APU generators are not interchangeable with the engine units. The GCU regulates the generator output and controls the starter/generator contactor. The GCU is located under the cabin floor near the AC GCUs. Retain interchangeable with engine

GCUs.

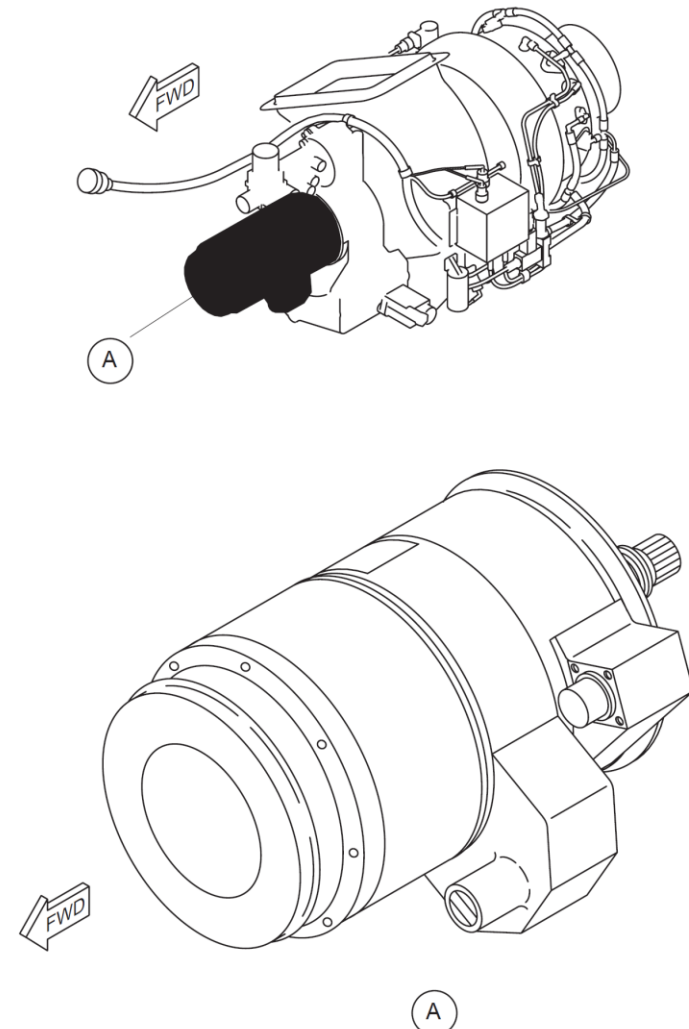


Figure 49-25 APU Starter/Generator

Ignition Exciter

[Refer Figure 49-26](#)

The ignition exciter is installed on the left side of the engine. The exciter is a capacitor discharge unit that uses 28 Vdc to give intermittent high voltage to the igniter plugs.

The exciter gives high voltage outputs and can be dangerous. When replacing the ignition exciter, be sure electrical power is removed from the exciter for a minimum of 5 minutes. Disconnect igniter cables from the igniter plugs. Discharge the exciter by touching the igniter cable ends to the igniter plug shells. Be sure exciter is fully discharged before removing the igniter cables and exciter.

Igniter Plugs

Two igniter plugs are installed, one on the left side and one on the right side of the engine. The igniter plugs are threaded into the engine combustor housing. The 10 millimeter, gap type igniters give a high voltage intermittent spark that ignites the fuel-air mixture in the combustor. The igniters are connected to the Ignition Exciter by two separate shielded ignition cables.

LEGEND

1. Exciter.
2. Upper Ignition Cable.
3. Upper Ignitor.
4. Lower Ignitor.
5. Lower Ignition Cable.

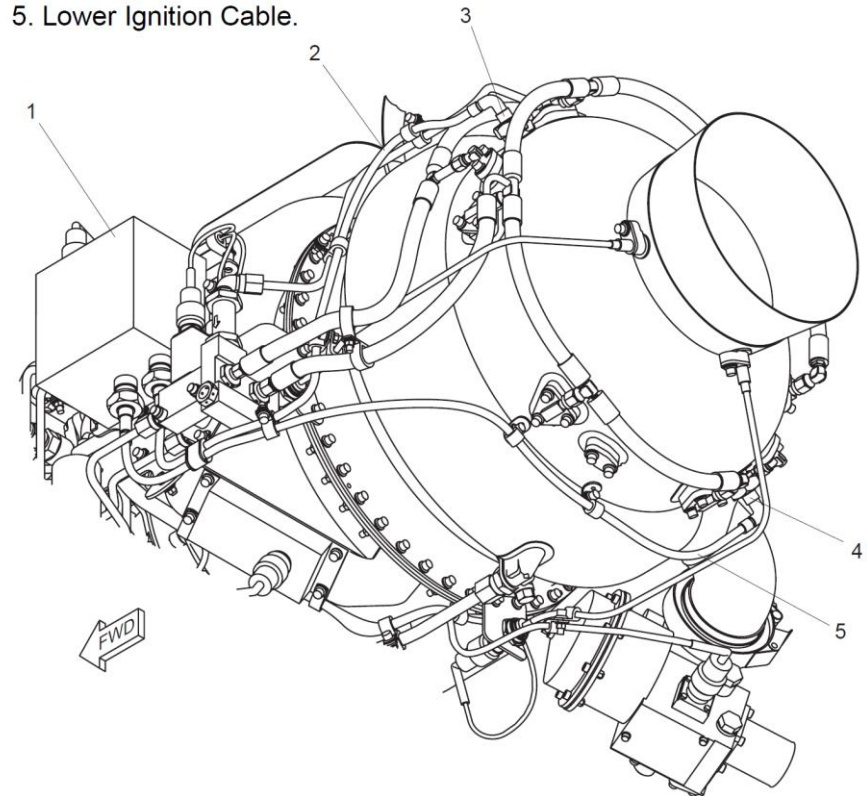


Figure 49-26 APU Ignition System

APU Start Sequence

[Refer Figure 49-27](#)

The start sequence begins with the momentary start command signal from the APU CONTROL panel. This opens the fuel servo valve.

The start fuel valve is energized open at 3% gas generator (Ng) speed which allows fuel flow to the two start nozzles and is ignited by the two ignitors.

When 25% Ng speed is reached the main fuel valve is energized open which allows fuel flow to the six main fuel nozzles. A flow divider valve in the fuel manifold assembly opens when fuel pressure reaches 300 psid (2068.4 kPad). This makes sure proper fuel pressure is maintained in the start nozzles.

The start fuel valve is de-energized closed at 70% Ng. When the start valve closes, P3 bleed air opens the Purge Check valve which blows residual fuel from the start fuel manifold and nozzles.

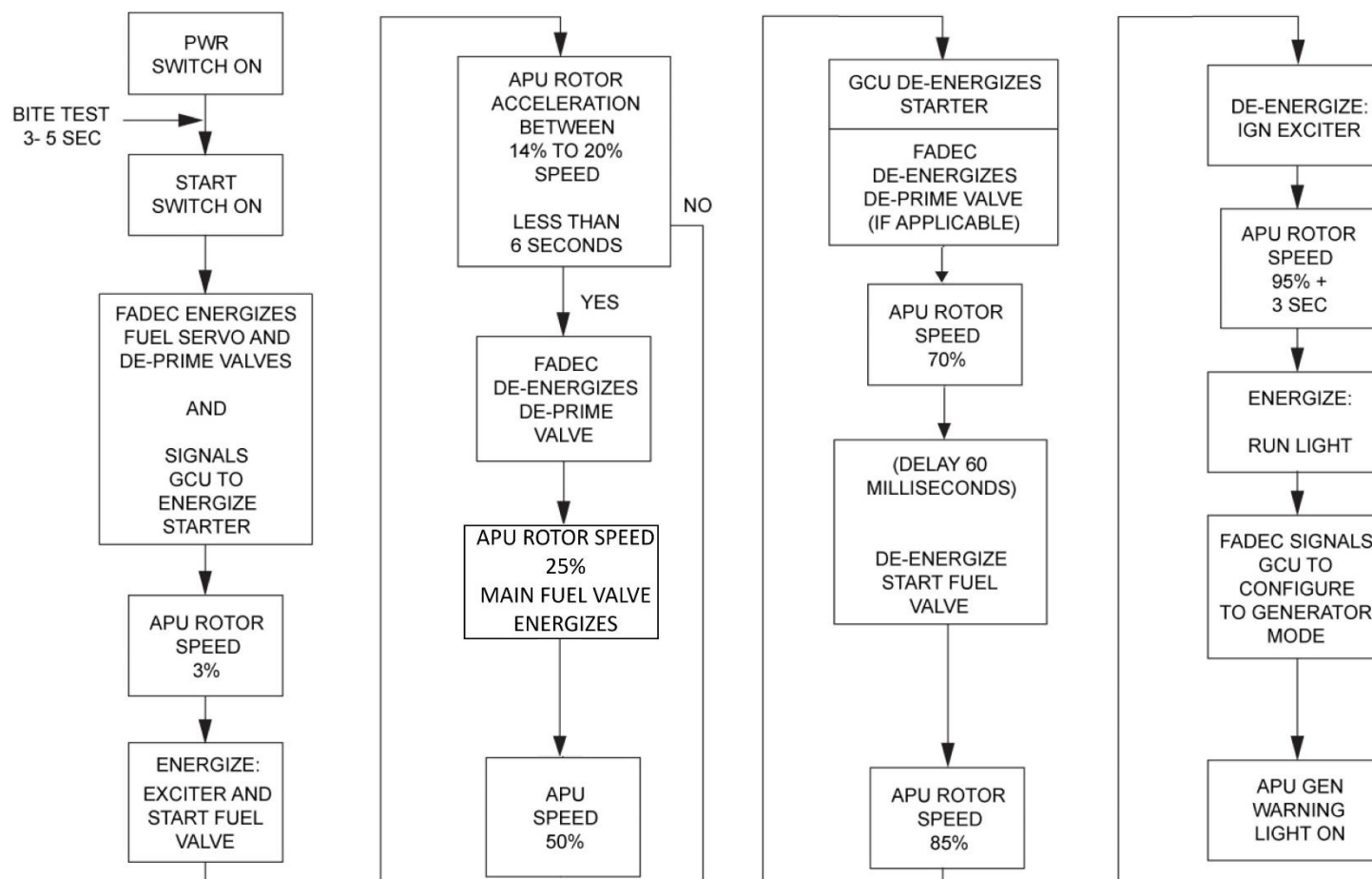


Figure 49-27 APU Start Sequence

APU Shutdown Sequence

Refer Figure 49-28

1. FADEC generated auto shutdown:

- Illogical switch input (example: APU ON switch pushed while either Weight Off Wheels (WOFW) signal from the PSEU is present)
- Airborne operation
- Low oil pressure
- Both speed signals failed
- Overspeed
- Underspeed
- Both EGT probes fail
- High EGT
- Main fuel valve failed (open, short, stuck)
- High oil temperature

- FADEC internal timer fail.

2. Aircraft generated auto shutdown:

- APU fire detection system
- APU fire bottle low pressure
- Generator overheat
- #1 WOW indicating air mode
- #2 WOW indicating air mode.

At any time that a major failure is detected by the FADEC, the APU is automatically shutdown. The 'fault' signal closes the fuel valves and illuminates the APU Caution light in the flight compartment, the WARN annunciation on the APU control panel, and extinguishes the RUN light.

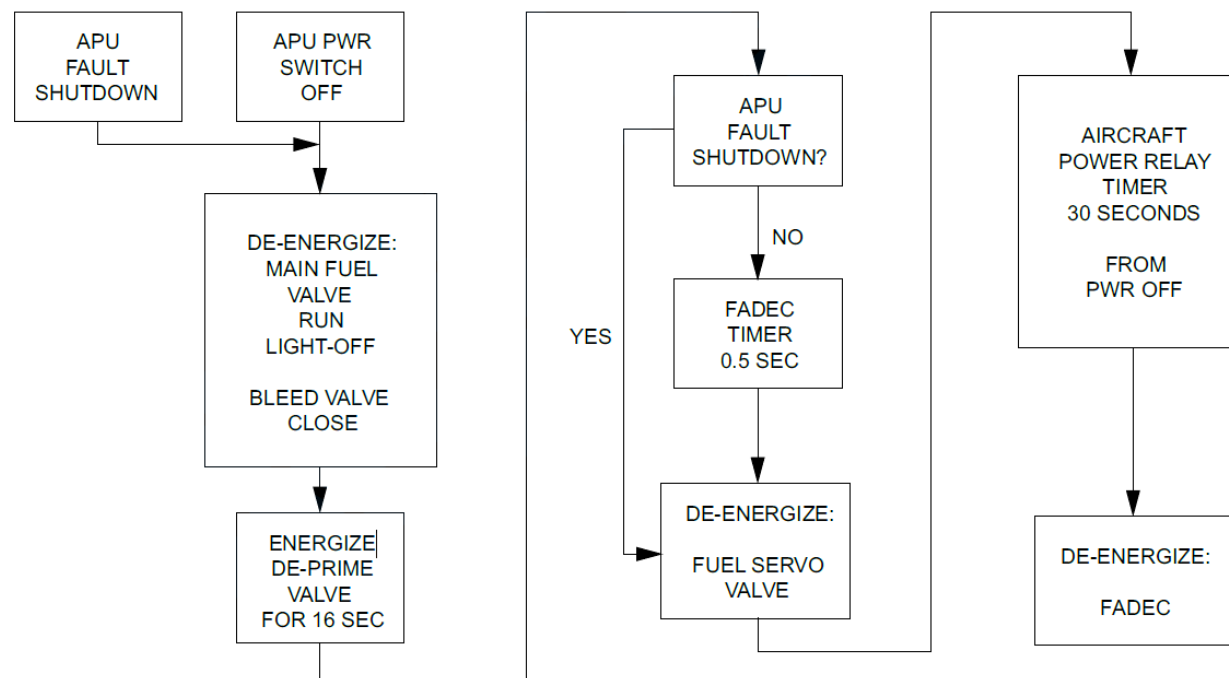


Figure 49-28 APU Shutdown Sequence

APU Fuel Valve Control – Electrical Schematic

[Refer Figure 49-29](#)

The start and ignition sequence is automatically controlled by the APU FADEC.

Selection of the PWR switchlight sends a power to the FADEC which enables the APU functions. Pressing the START switchlight starts the ignition sequence.

For the start the FADEC does the following:

- The APU Fuel SOV is energized open.
- The Fuel Servo valve is modulated open.
- The Ignitors are energized. The capacitor ignition exciter receives 28VDC and sends out intermittent high voltage to the ignitor plugs. The gap type ignitors give intermittent sparks which ignites the fuel-air mixture in the combustion chamber.
- The Start Fuel valve is energized open at 3% NG.
- The Oil De-prime valve is energized closed.
- Sends a 'Starter Operate' signal to the GCU connector.

The APU GCU processes the starter operate signal and sends a 28VDC Start Control signal to the DC contactor box to energize the relay K26, in the contactor box.

The relay K26 sends 28 VDC from the right main bus to the APU Starter/Generator terminal. This causes the 'starter' to rotate the APU up to 50% NG.

The APU 50% speed signal to the CGU causes the GCU to remove the 28 VDC from the relay K26 and de-energize the 'starter'.

At 50% NG, the engine speed the de-prime valve is de-energized open.

When the APU GCU senses the NG is 95% plus 3 seconds, the generator function is available which is indicated as a WARN light on the APU CONTROL panel.

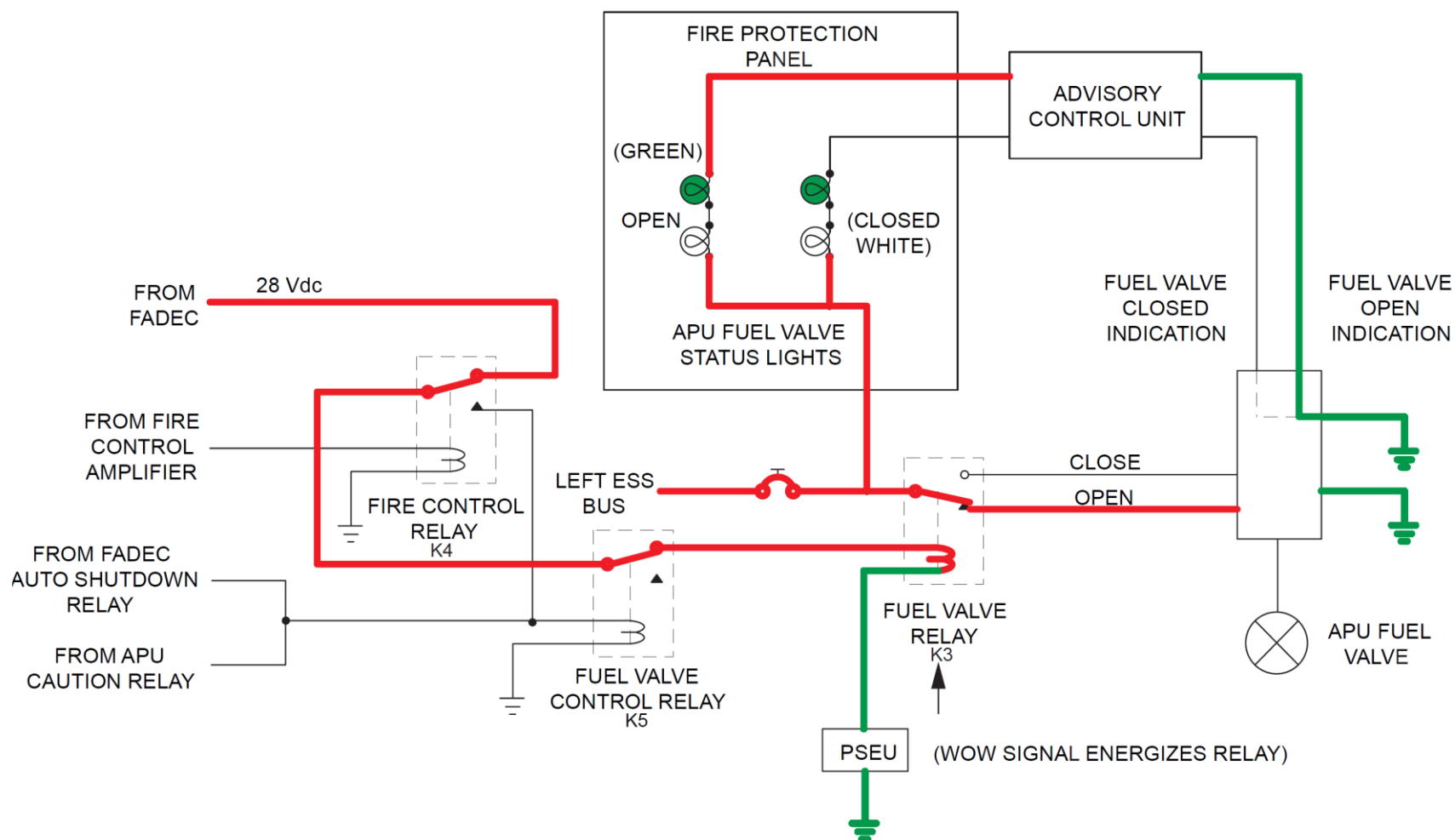


Figure 49-29 APU Fuel Valve Control – Electrical Schematic

Air System

Introduction

The APU bleed air system supplies compressed air for air conditioning and other aircraft pneumatic requirements.

General Description

The aircraft air inlet supplies the air to the APU air inlet assembly that is then compressed by the APU compressor. The compressed air is used in the operation of the APU and to supply air to the aircraft pneumatic systems.

Detailed Description

Refer Figure 49-30

The APU bleed air system supplies compressed air for air conditioning and other aircraft pneumatic requirements.

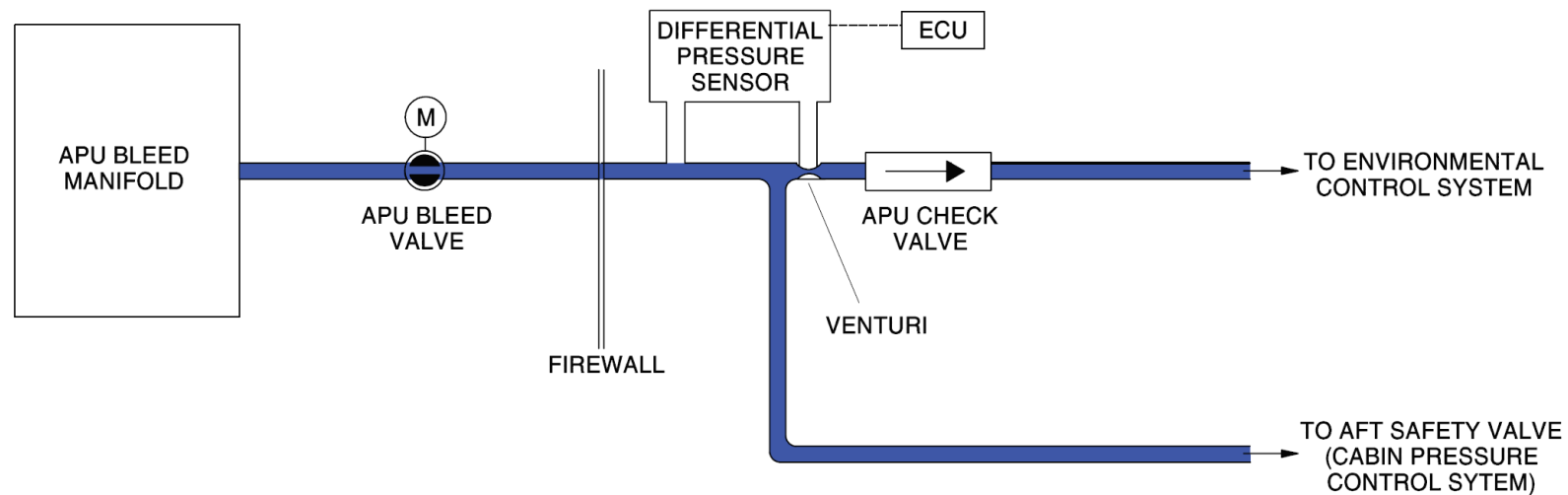


Figure 49-30 APU Bleed Air Schematic

Air System – Component Description

APU Bleed Valve

[Refer Figure 49-31](#)

The bleed valve is an electro-pneumatic butterfly valve that is normally closed. The valve is located on the APU combustor housing bleed air duct and controlled by the FADEC. FADEC receives a signal from the BL AIR switch on the APU control panel. The FADEC uses signals from the APU exhaust thermocouples to control the Bleed Valve during bleed air delivery. If the APU exhaust gas temperature (EGT) reaches 692 °C the Bleed Valve will be modulated to reduce bleed air flow. This gives maximum air flow delivery without exceeding the APU exhaust gas temperature limits. Modulating bleed air flow in response to exhaust gas temperature limits lets generator loads have priority over bleed air delivery.

Bleed Air Duct

The bleed duct directs bleed air to the ECS. It is installed on the right side of the auxiliary power unit (APU).

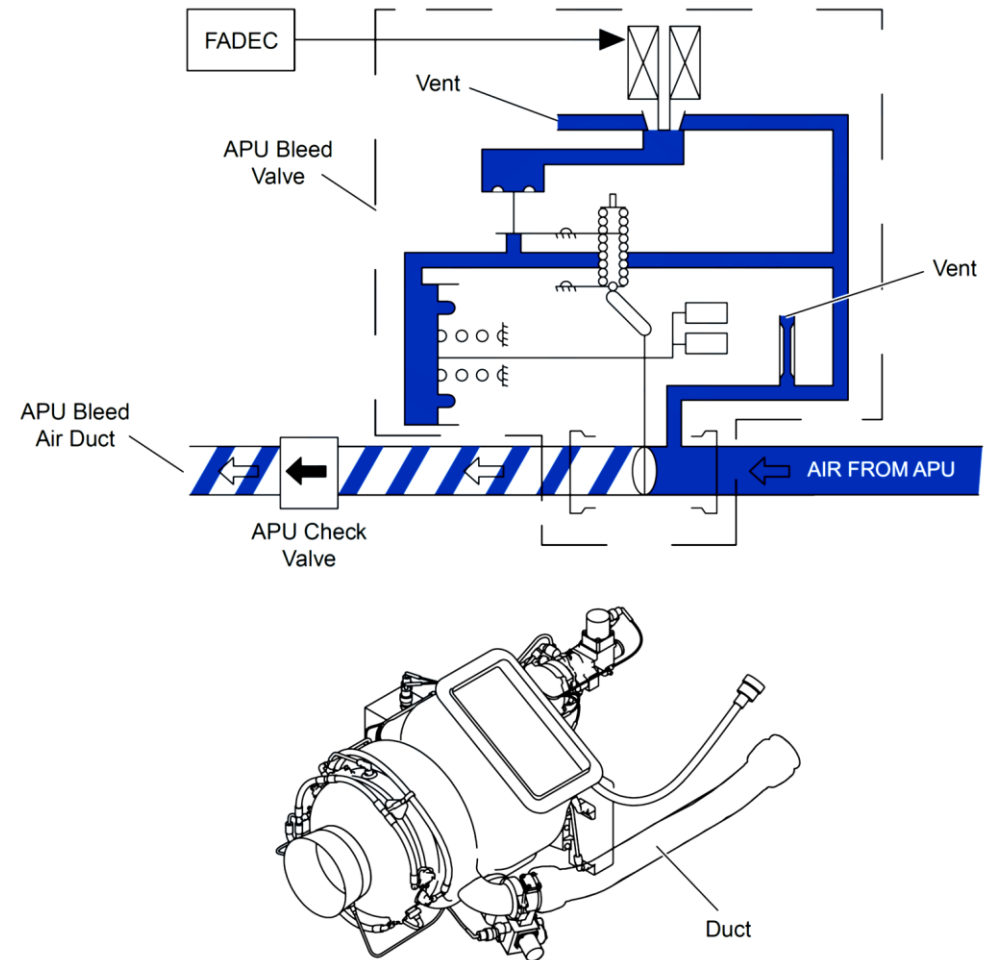


Figure 49-31 Bleed Valve and Duct

APU Bleed Request – Electrical Schematic

[Refer Figure 49-32](#)

The normally closed bleed valve is controlled by the FADEC.

- Selection of the BL AIR switchlight to OPEN on the APU CONTROL panel sends 28 VDC from the APU CONTROL panel to the APU relay panel.
- The 28 VDC goes through relay K11 and K6 to the FADEC as a 'Bleed Air Request'. This causes the FADEC to energize the Bleed air valve open.
- If either Bleed Duct switch senses a duct air leak, relay K11 will close and remove the Bleed Air Request signal. Auxiliary contacts in K11 will latch the relay.
- To reset the system a selection of the Bleed Air switch to OFF then back to ON is required.
- If either engine Bleed switch is selected to the on position, relay K6 will close and remove the Bleed Air Request signal.
- The loss of the Bleed Air Request signal causes the FADEC to close the Bleed Air valve.

The FADEC also receives signals from the APU exhaust thermocouples for control of the Bleed Valve during bleed air delivery. As the APU EGT reaches 692°C the FADEC will modulate the Bleed Valve toward closed. This logic supplies bleed air without exceeding the APU EGT limits. Modulating bleed air flow in response to EGT limits gives the generator priority over bleed air.

The APU check valve in the bleed air duct is held open by the bleed air coming from the APU but stops the air from the PACKS to prevent reverse flow into the APU engine bleed air.

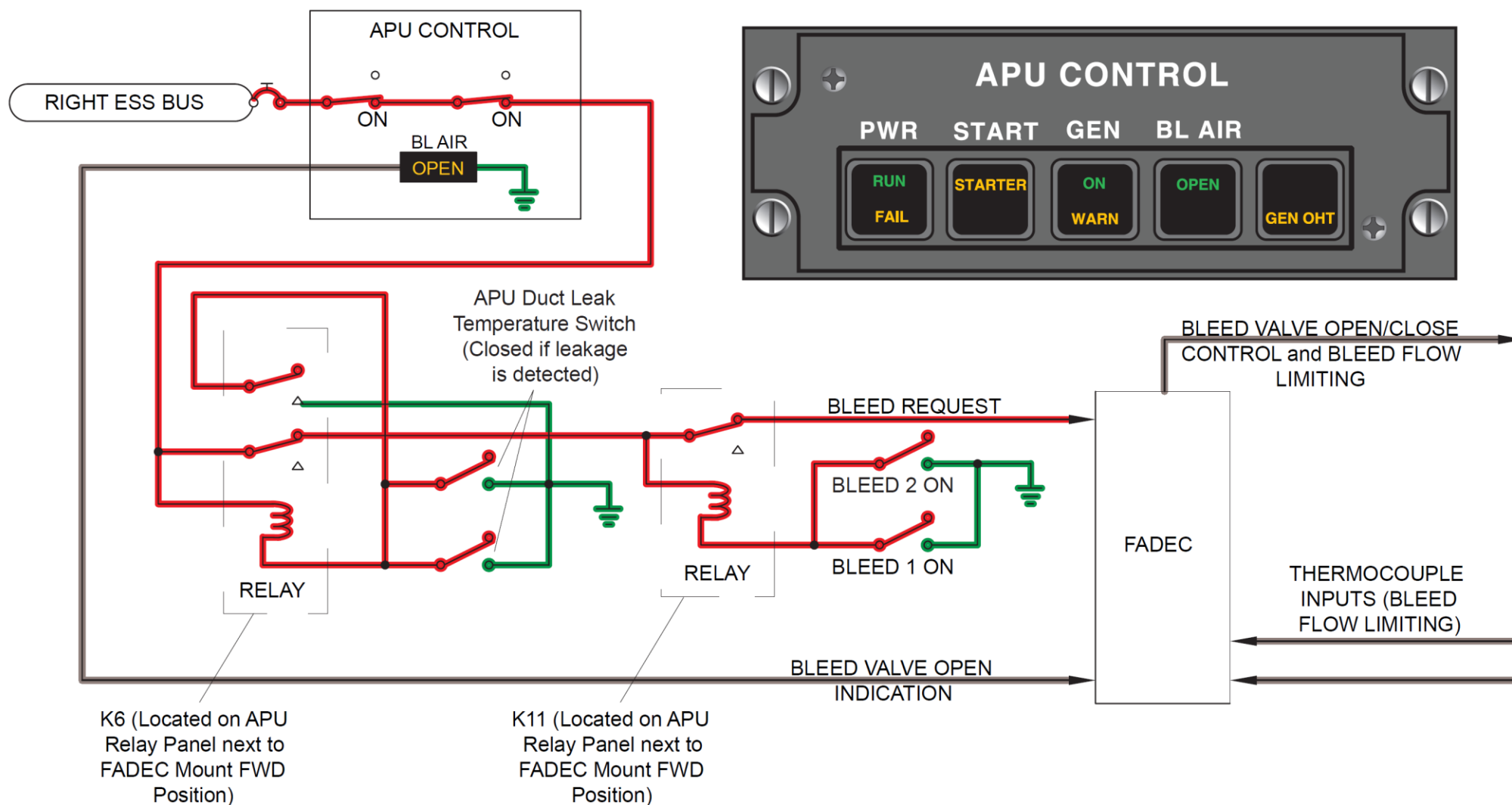


Figure 49-32 APU Bleed Request – Electrical Schematic

Indicating System

Introduction

The Indicating System supplies the engine operating parameters to the Full Authority Digital Electronic Control (FADEC) for APU engine monitoring and control.

General Description

The Indicating System has the components that follow:

- Sensor, Ambient Pressure
- Sensor, Speed
- Thermocouple
- Switch, High Oil Temperature
- Switch, Low Oil Pressure
- Data Memory Module
- APU Fire Protection Panel

Detailed Description

At start initiation, the FADEC records the Exhaust Gas Temperature (EGT). This EGT value is used for reference during the start sequence where specific rises in EGT need to occur before the start sequence can continue. During the start sequence FADEC uses signals from the:

- Ambient pressure sensor
- Speed sensor
- Thermocouples,

to schedule fuel flow and initiate and monitor the start sequence.

Indicating System – Component Description

Sensor, Ambient Pressure

[Refer Figure 49-33](#)

The sensor is located at the left side of the APU. The sensor supplies ambient pressure information to the FADEC prior to start. This information is used for altitude compensation of the fuel during start and operation of the APU. A loss of the ambient pressure signal results in an advisory fault and the FADEC

will default to sea level ambient pressure.

Sensor, Speed

The sensor is installed to the air inlet housing. The sensor is a single-probe, dual coil device. The magnetic probe extends through the air inlet housing and aligns with the lobes on the quill shaft. The dual coils give redundant speed signals. The sensor generates redundant sinusoidal signals, which are transmitted to the FADEC and used for start sequencing and speed sensing (overspeed and underspeed). The two signals are averaged by FADEC to give the rated speed. If the difference between the two signals exceeds 5%, the FADEC will use the higher value. Failure of one signal will result in an advisory fault. Loss of both signals will cause an automatic APU shutdown.

Thermocouple

The two single-element, closed end, chromel/alumel thermocouples are located on the APU exhaust flange and extend into the exhaust gas stream. The thermocouples are installed 90 degrees apart, at the five and eight o'clock positions and supply millivolt signals that are used by FADEC for exhaust gas temperature sensing. The FADEC averages the two signals to give the exhaust gas temperature. If the difference between the two thermocouples is greater than 200 °F (93.3 °C), the FADEC will use the higher value. An open or short failure of one thermocouple will result in an advisory message. Failure of both thermocouples will cause an APU shutdown.

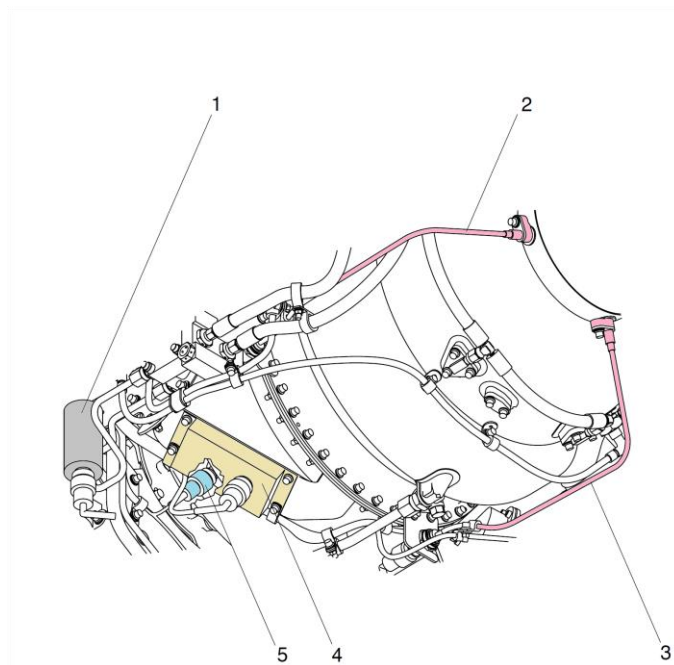
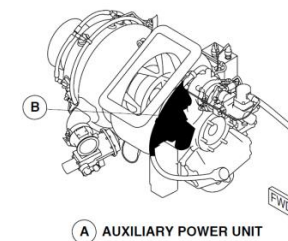
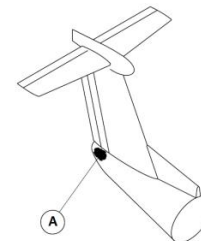


Figure 49-33 APU Sensors Detail

LEGEND

1. Ambient Pressure S
2. Thermocouple 1.
3. Thermocouple 2.
4. Data Memory Modul
5. Speed Sensor.



LEGEND

1. High Oil Temperature Switch.
2. Low Oil Pressure Switch.

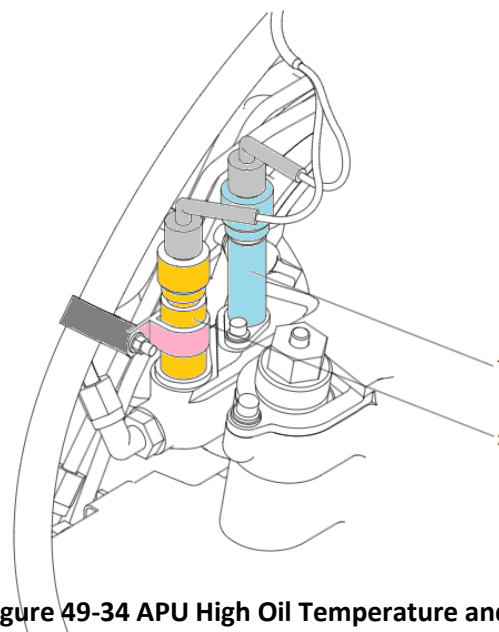


Figure 49-34 APU High Oil Temperature and Low Oil Pressure Switches

Switch, High Oil Temperature

[Refer Figure 49-34](#)

The switch is located on the forward face of the gearbox assembly. The switch is normally open and closes if the oil temperature exceeds 275 °F (135 °C). A closed switch will result in the FADEC shutting down the APU.

Switch, Low Oil Pressure

[Refer Figure 49-34](#)

The switch is located on the forward face of the gearbox assembly and senses oil pressure, downstream of the oil filter. The normally closed switch opens when oil pressure exceeds 35 psig (241 kPag), the switch will close when oil pressure decreases below this value during APU operation. Prior to start, the FADEC checks the position of the switch. During operation the FADEC begins checking for low oil pressure after Ready To Load (RTL). An open switch prior to start (failed switch), or a closed switch during operation (failed switch or low pressure) will result in the FADEC automatically shutting down the APU.

Data Memory Module

[Refer Figure 49-33](#)

The Data Memory Module (DMM) is a solid state device located on the engine assembly. The DMM contains a mirror image of the data contained in the non-volatile memory (NVM) within the FADEC. Data stored in the DMM includes:

- APU operating hours and starts
- APU and FADEC fault history, 50 shutdown and 50 advisory faults
- FADEC, APU and DMM serial numbers.

FADEC updates the DMM at the same time the data is updated in-the FADEC NVM. This feature enables hours/cycles and fault history-to be resident on the engine assembly.

The DMM can communicate directly with a laptop PC for-interrogation either on-wing or at a repair station. Should the FADEC-or DMM be replaced on-wing, the data is automatically copied to the-newly replaced item,

controlled by software without technician-intervention.

After ready to load, which occurs at 95% speed plus 3 seconds,-ambient pressure is used by FADEC to set the minimum and-maximum fuel set points. The speed sensor signal is used by-FADEC, to ensure 100% speed is maintained by increasing or-decreasing fuel flow through the fuel servo valve. The signal is also-used to monitor for overspeed and underspeed conditions.

During operation FADEC also monitors the:

- Exhaust gas temperature
- Oil pressure
- Oil temperature

The conditions that follow will cause an automatic APU shutdown:

- Low oil pressure below 35 psig (241 kPag)
- High oil temperature above 275 °F (135 °C)
- Overtemperature EGT above 1890 °F (1032 °C) during start, 1324°F (718 °C) during run
- Underspeed less than 95%
- Overspeed 105%

APU Fire Protection Panel

[Refer Figure 49-35](#)

- **1** - VALVE CLOSED Light is on when the APU fuel shut-off valve is closed.
- **2** - FUEL OPEN Light is on when the APU fuel shut-off valve is open.
- **3** - EXTG (guarded) Switch Light used to operate the fire extinguisher. This switch overrides the timer unit.
- **4** - BTL LOW Light comes on when the fire extinguishing bottle loses pressure.
- **5** - FAULT Light comes on when the Control Amplifier detects a fault in the fire detection system.
- **6** - BTL ARM Light is on when fire extinguishing bottle is ready to fire.
- **7** - FIRE Light will come on when a APU fire is detected.
- **8** - FIRE TEST Switch Starts the test sequence for the fire detection and extinguishing systems for the APU.

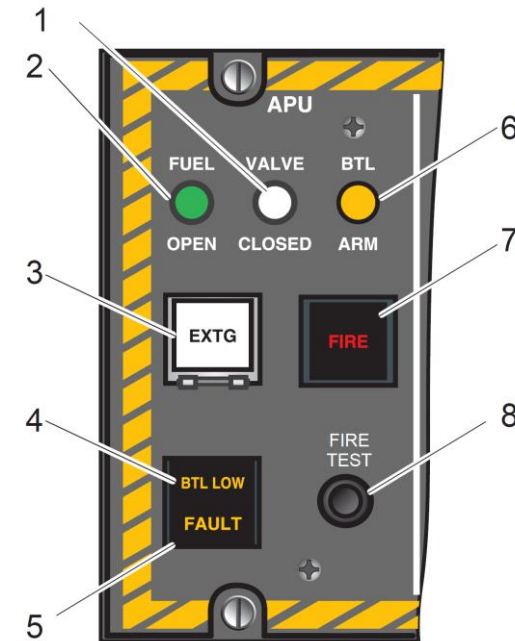


Figure 49-35 APU Fire Protection Panel

APU Control – Electrical Schematic

[Refer Figures 49-36 \(Sheet 1 of 2\)](#)

The APU FADEC provides control and indication and records APU data.

PWR Switchlight is Pushed

- This arms the APU start circuits and opens the APU fuel valve
- FUEL OPEN green light comes on.

START Switchlight is Pushed

- This energizes APU start sequence
- STARTER amber light comes on
- At 50% Ng the STARTER light will go off
- After APU reaches 95% for 3 seconds, RUN green light comes on
- APU generator is in standby
- WARN amber light comes on.

GEN WARN Switchlight is Pushed

- The APU generator sends 28 VDC to the right main DC bus
- ON green light comes on.
-

BL AIR Switchlight is Pushed

- APU bleed air valve opens
- OPEN green light comes on.

FAIL Amber Light Comes On

- An APU failure is detected and the FADEC shuts down the APU automatically
- APU bleed air valve is closed
- APU fuel valve is closed
- VALVE CLOSED white light comes on
- APU amber caution light comes on.

GEN OHT Amber Light Comes On

- The APU DC starter / generator temperature is more than 166°C (330°F)
- An APU failure is detected and the FADEC shuts down the APU automatically
- APU bleed air valve is closed
- APU fuel valve is closed
- VALVE CLOSED white light comes on
- APU amber caution light comes on.

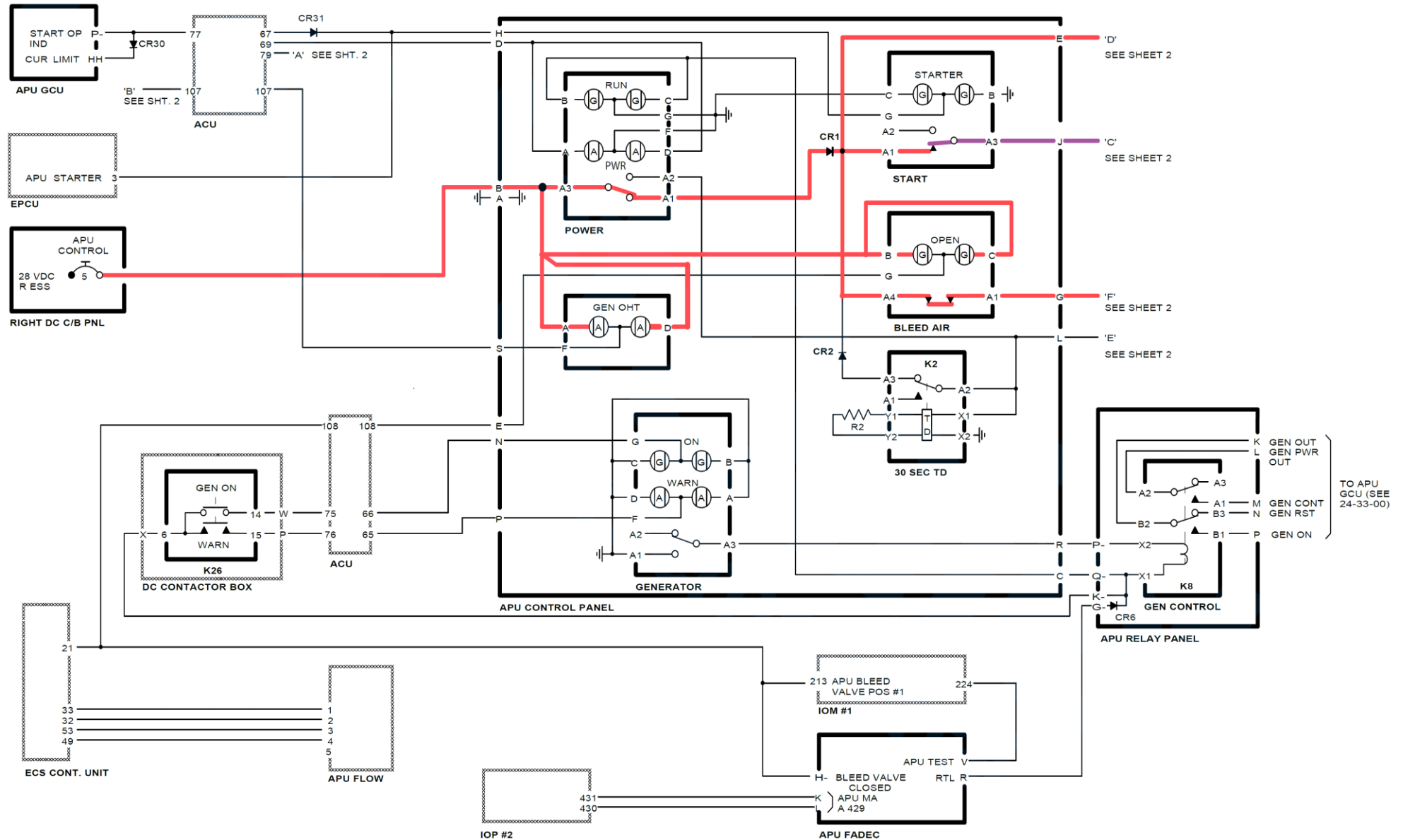


Figure 49-36 (Sheet 1 of 2) APU Control – Electrical Schematic

[Refer Figure 49-37 \(Sheet 2 of 2\)](#)

During start initiation, the FADEC uses EGT thermocouple for reference temperature to sense a rise in EGT before the start sequence can continue. The FADEC uses the speed sensor data for fuel to sequence the fuel valves and ignition.

The Speed sensor dual coils give redundant speed signals to the FADEC and for the start sequence, over-speed and under-speed sensing. The FADEC creates an average of the two signals to create the rated speed.

If the two signals difference exceeds 5%, the FADEC will use the higher value.

One signal fail will cause an advisory message found in the CDS.

Both signals fail will cause the FADEC to automatically shutdown the APU.

Two single element thermocouples supply a millivolt signal to the FADEC for exhaust gas temperature sensing. The FADEC averages the two signals for EGT sensing.

If the two signals difference exceeds 200°F (93.3 °C), the FADEC will use the higher value.

One signal fail will cause an advisory message found in the CDS.

Both signals fail will cause the FADEC to automatically shut down the APU.

The High Oil Temperature normally open switch closes if the oil temperature exceeds 275°F (135°C).

A closed switch will cause FADEC to automatically shutdown the APU.

The Low Oil Pressure normally closed switch senses the oil pressure downstream of the oil filter and opens when oil pressure exceeds 35 psig.

An open switch prior to start (failed open), or a closed switch during operation will cause the FADEC to automatically shut down the APU.

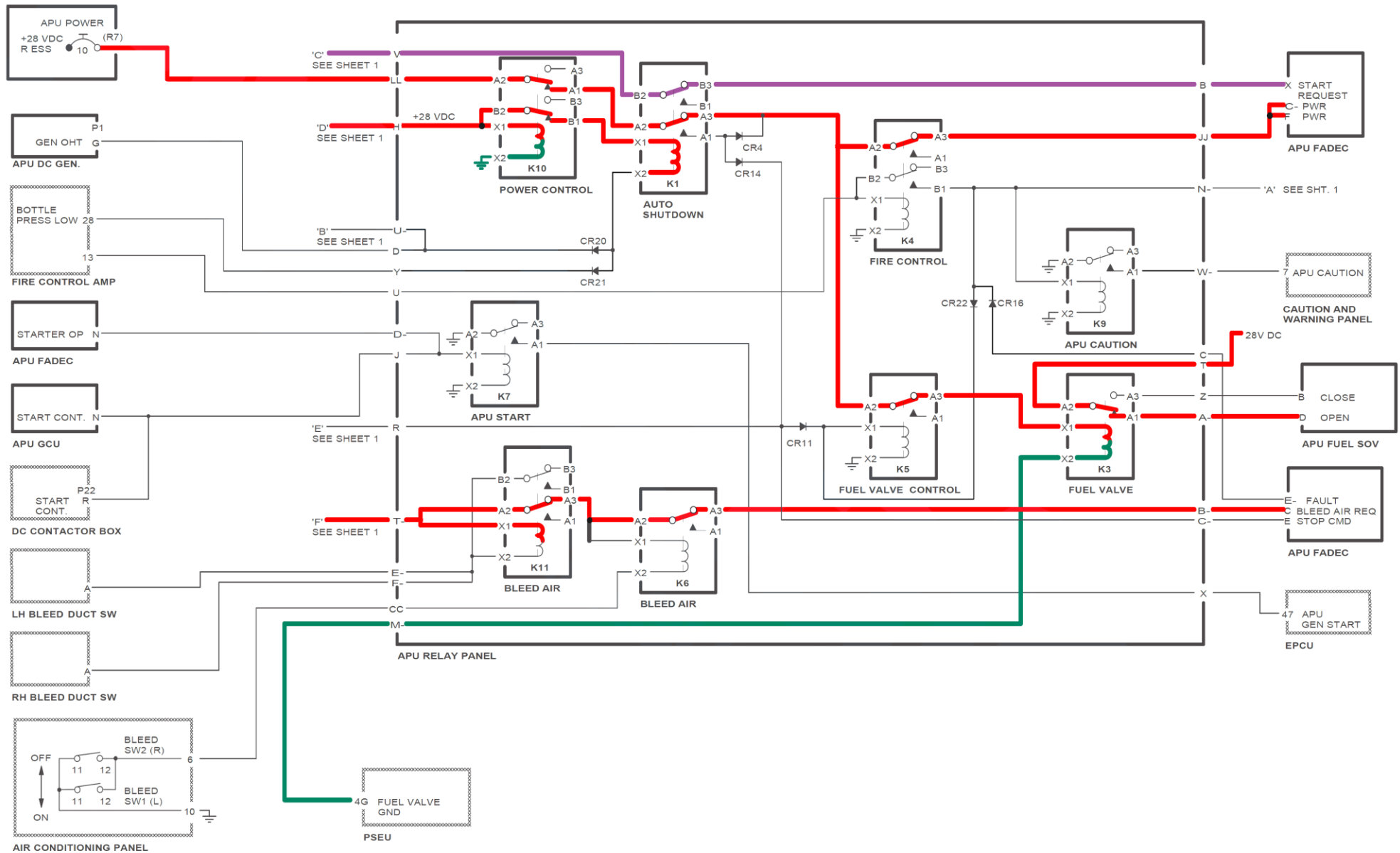


Figure 49-37 (Sheet 2 of 2) APU Control – Electrical Schematic

Fault Diagnostics

Centralized Diagnostic System

The Centralized Diagnostic System (CDS) allows access to electronic APU fault detection and fault storage data. The CDS displays the fault codes operational testing and troubleshooting.

[Refer Figure 49-38](#)

Central Diagnostic System Unit

The Central Diagnostic System Unit interfaces with the following:

- APU FADEC
- Aircraft avionics
- ARCDU 1
- ARCDU 2
- Maintenance panel
- ACARS (optional).

Downloading FADEC Data

Set the APU PWR switch to the ON position. The switch must stay ON for the duration of the Smart Terminal inquiry. Connect the Smart Terminal cable between the laptop and the J3 connector on the FADEC. Start the Smart Terminal to download the parametric data for troubleshooting. Follow the on-screen instructions for operation of the troubleshooter.

Audio and Radio Control and Display Units (ARCDU)

The ARCDU 1 or ARCDU 2 can display the CDS menus which includes the APU fault information sent by the FADEC to the CDS unit.

Operating the ARCDU

The ARCDU has Selection keys for selecting the items displayed on the screen.

Activation of the CDS (for APU maintenance functions) requires the following:

- Aircraft on the ground (WOW)
- Airspeed less than 50 knots
- The CDSGNDMAINT switch set to ON (located on the aircraft maintenance panel)

- The 'MAINT' key pushed on ARCDU 1 or ARCDU 2
- APU PWR selected ON.

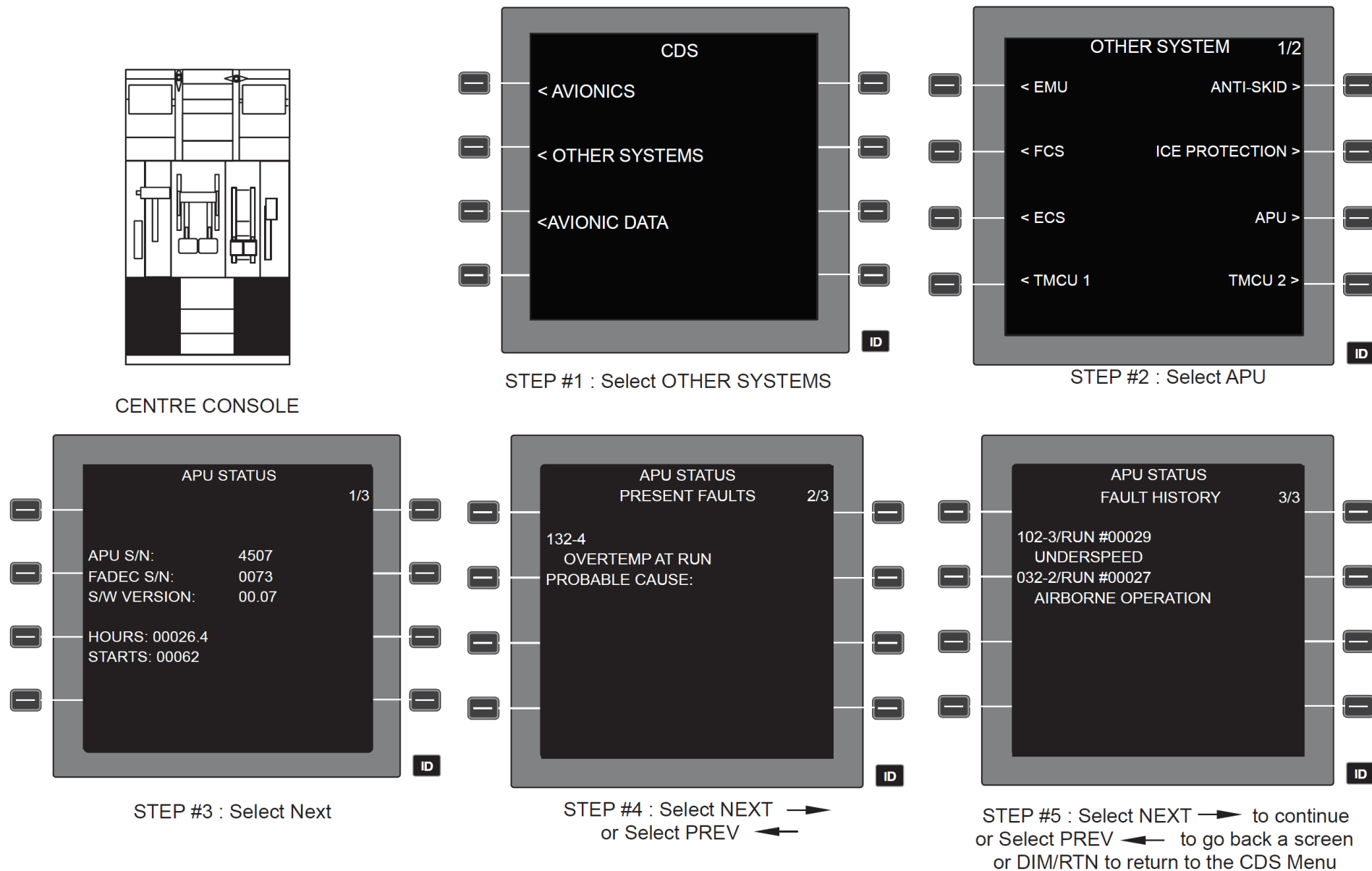
APU Status Pages

There are three types of APU STATUS pages, titled and numbered as follows:

- APU STATUS — 1/1
- APU STATUS PRESENT FAULTS
- APU STATUS FAULT HISTORY

The ARCDU APU STATUS pages provides the following information:

- APU Serial Number
- FADEC Serial Number
- Software Version Number
- APU Hours
- APU Starts.


Figure 49-38 APU CDS Pages

Oil System

Introduction

The APU oil system supplies pressurized oil to the gearbox and the engine rotor assembly to lubricate and cool the gears and the bearings.

General Description

[Refer Figure 49-39](#)

A positive displacement gear type oil pump is located inside the gearbox and supplies oil flow to the system. The oil pump supplies oil to the gearbox and the engine rotor assembly through the fuel/oil cooler where the oil is cooled by fuel. If oil flow is restricted through the cooler, a cooler bypass valve opens at 65 to 70 psig (448 to 483 kPag). The valve is located in the gearbox. The oil is filtered through a 10 micron replaceable filter and then flows into the oil system.

The oil pressure relief valve, located between the oil filter and the gearbox oil jets, maintains the system oil pressure of 60 psig (414 kPag). The valve is a non-adjustable component.

High oil temperature and low oil pressure switches monitor the oil system. If either low oil pressure or high oil temperature condition occurs, the APU will shutdown.

The gears and bearings are lubricated and cooled by oil jets and oil mist that is generated by the rotating gears. The oil/air separator is part of the fuel control gear and separates the oil from the air inside the gearbox. The oil returns by gravity to the oil sump and the air is vented overboard through the drain mast vent.

Lubrication and cooling of the rotor bearings is supplied by the gearbox oil pressure system. Oil is injected into the hollow pinion shaft that is splined into the rotor shaft. The pinion shaft delivers oil to the passages located inside the rotor shaft. During engine operation, centrifugal force lets the oil flow through the internal passages to the ball and roller bearings. Oil return from the rotor bearings is accomplished by gravity flow to the gearbox oil sump.

The oil system contains the components that follow:

- Gear type pump
- De-prime valve
- Fuel/oil cooler
- Fuel/oil cooler bypass valve
- Filter
- Pressure relief valve
- High temperature switch
- Low pressure switch
- Oil/air separator
- Carbon seal drain
- Labyrinth seal
- Oil filler port
- Sight glass
- Magnetic drain plug

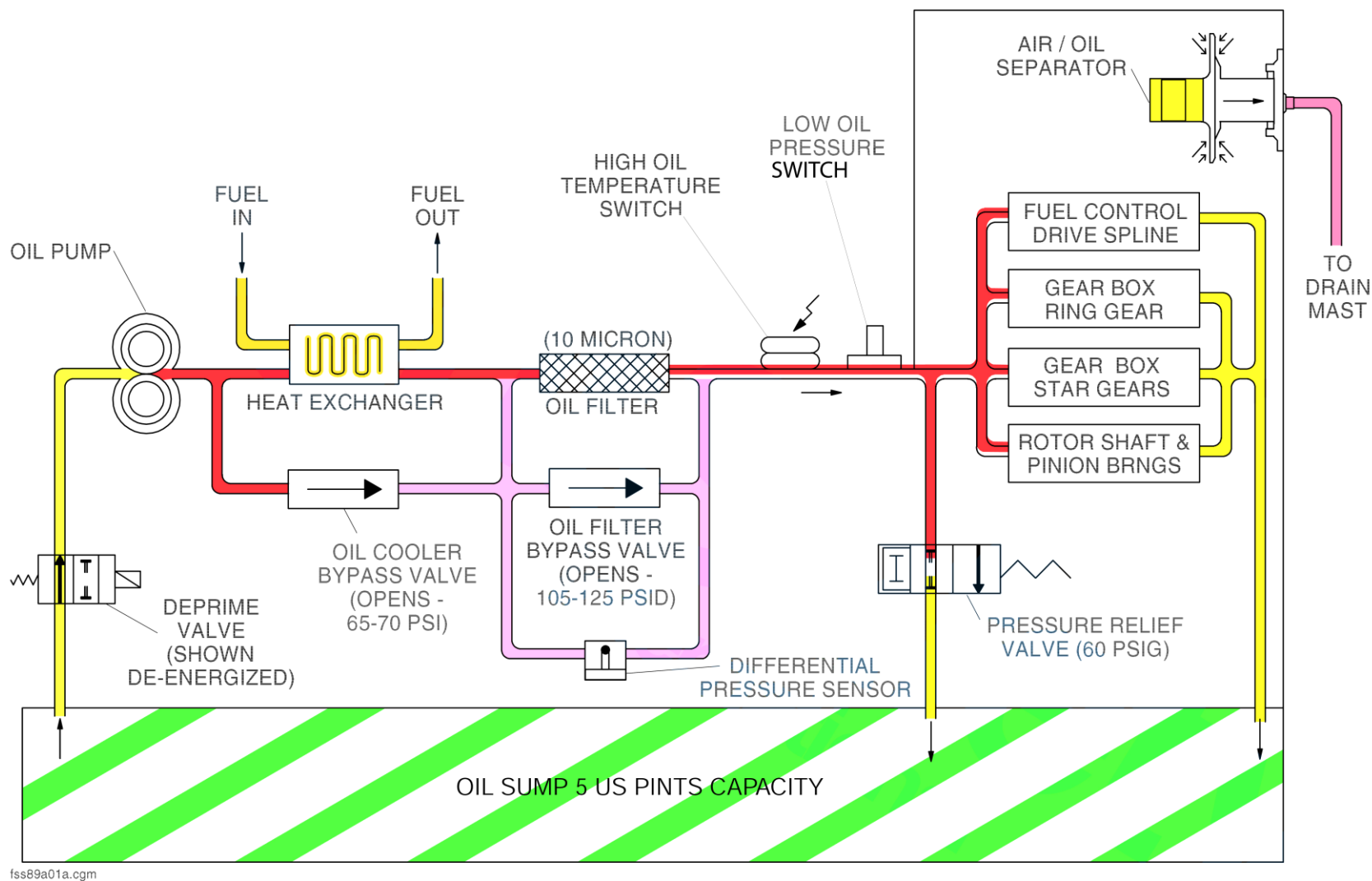


Figure 49-39 APU Oil System Schematic

Oil System – Component Description

[Refer Figure 49-40](#)

Oil Pump

A positive displacement gear type oil pump is located inside the gearbox and supplies oil flow to the system.

Deprime Valve

The deprime valve is a solenoid operated valve and is located on the lower, right side of the gearbox. The valve is energized open to prevent the oil pump from supplying oil during engine startup and shutdown. Above 50% engine speed the deprime valve is de-energized closed and the oil pump produces oil flow. During engine shutdown the deprime valve is energized open for 16 seconds, to prevent oil pump output and to let the oil remaining in the system

drain back into the sump.

Fuel/Oil Cooler

The Fuel/Oil cooler is located on the lower, right side of the gearbox. The cooler is a fuel/oil heat exchanger that uses fuel from the fuel control to supply cooling of the oil.

Fuel/Oil Cooler Bypass Valve

The valve is located in the gearbox. In the event the oil flow is restricted through the cooler, the cooler bypass valve opens at 65 to 70 psig (448 to 483 kPag).

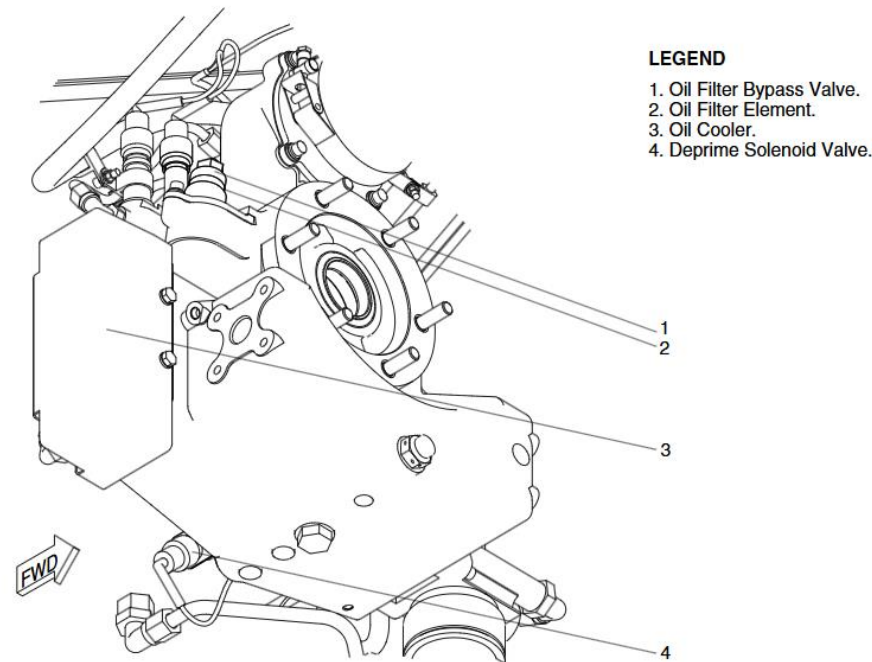


Figure 49-40 Oil System Components

Oil Filter

[Refer Figure 49-41](#)

Oil filtration is supplied by a disposable 10 micron filter. The oil filter bypass valve assembly includes a visual impending by pass indicator and a bypass valve. The indicator is visible when differential pressure across the filter reaches 60 to 80 psid (414 to 552 kPad). The bypass valve opens when the differential pressure across the filter reaches 105 to 125 psid (724 to 862 kPad).

Oil Pressure Relief Valve

The oil pressure relief valve, located between the oil filter and the gearbox oil jets, maintains the system oil pressure of 60 psig (414 kPag). The valve is a non-adjustable and non-replaceable component.

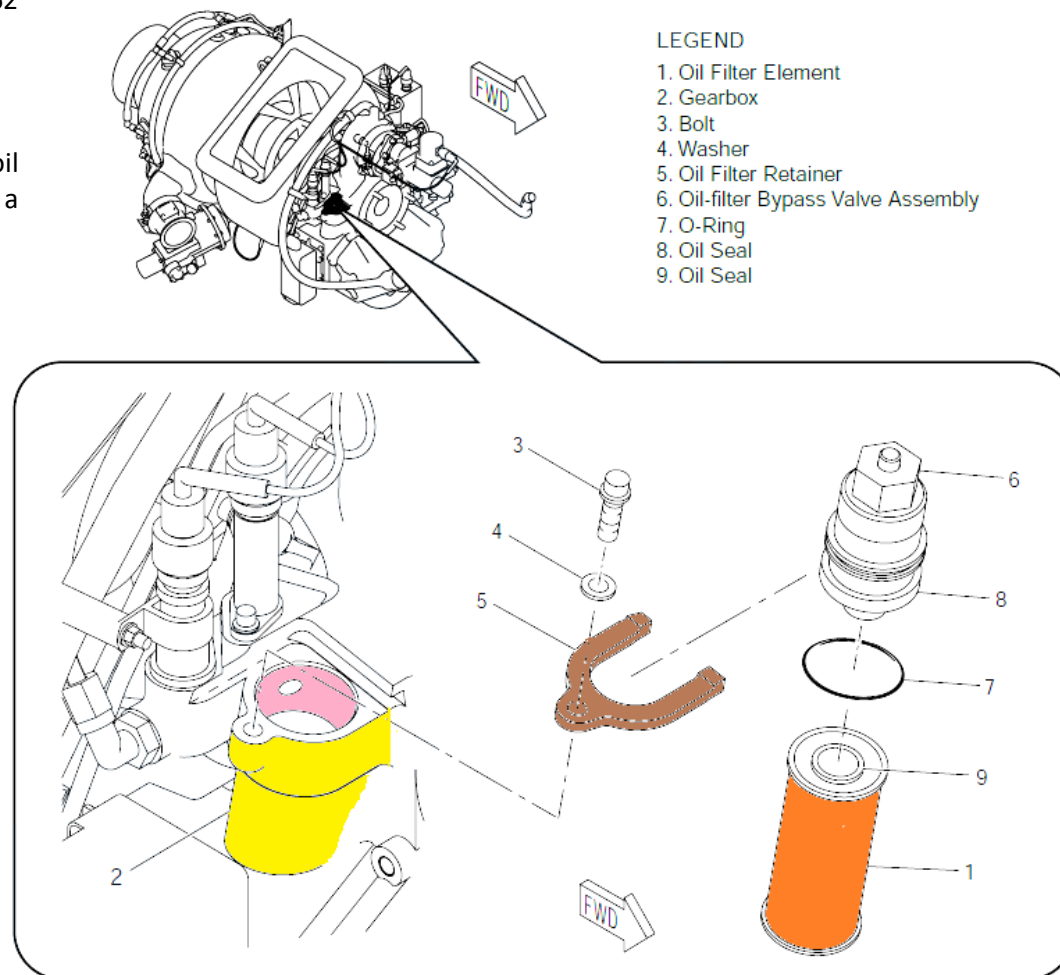


Figure 49-41 Oil Filter

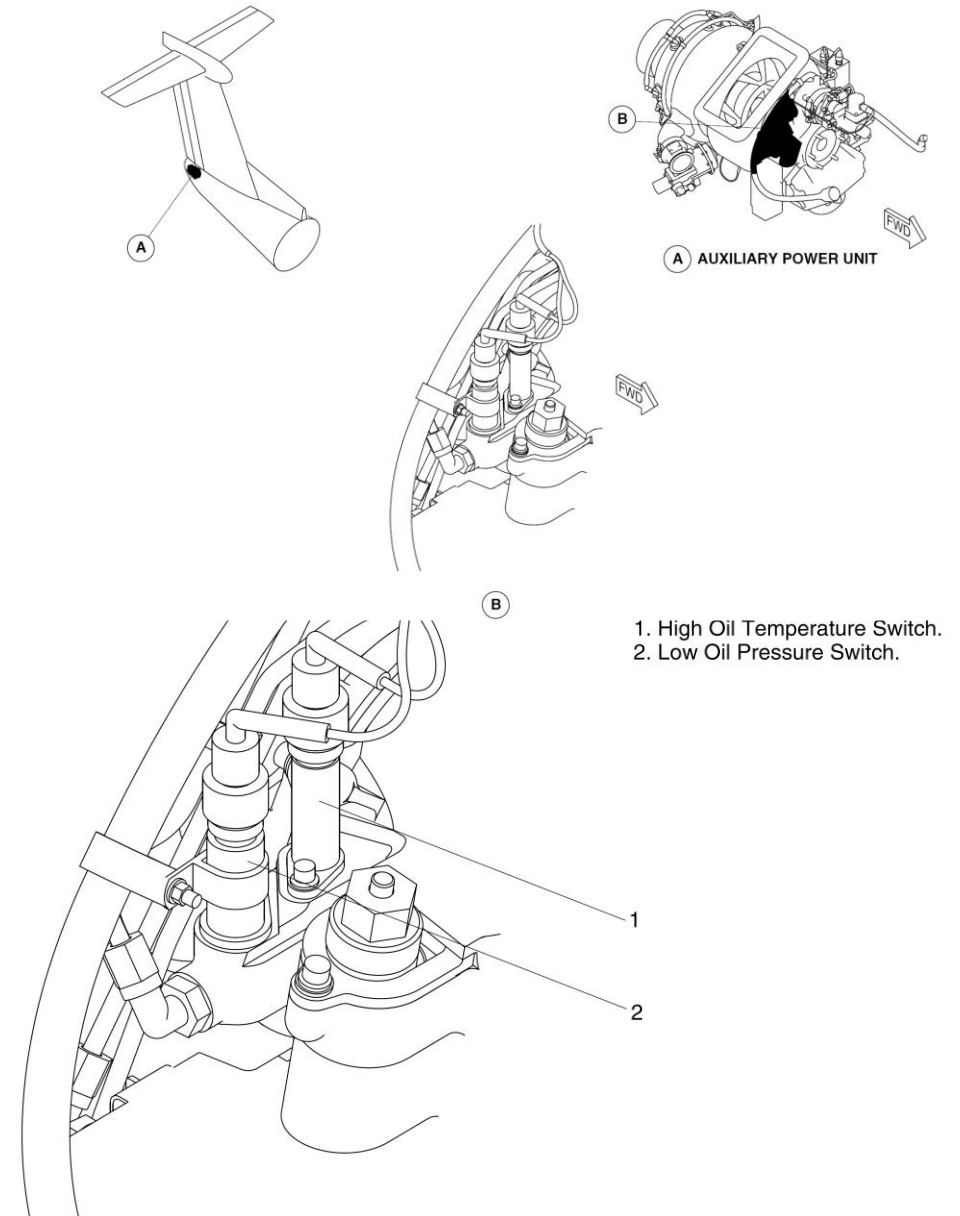
High Oil Temperature Switch

[Refer Figure 49-42](#)

The high oil temperature switch is a normally open switch and closes if the oil temperature is more than $275 \pm 5^\circ\text{F}$ ($135 \pm 3^\circ\text{C}$). A closed switch will result in the APU shutting down.

Low Oil Pressure Switch

The Low Oil pressure Switch senses oil pressure downstream of the oil filter. The normally closed switch opens when oil pressure is more than $35 \pm 5\text{ psig}$ ($241 \pm 34.5\text{ kPag}$) and remains open during normal operation of the APU. Oil pressure is not monitored until the APU has reached 100% speed and the switch has been closed for 12 seconds. A closed switch will initiate a shutdown of the APU.



1. High Oil Temperature Switch.
2. Low Oil Pressure Switch.

Figure 49-42 APU High Oil Temperature and Low Oil Pressure Switches

Oil Air Separator

The oil/air separator is part of the fuel control gear and separates the oil from the air inside the gearbox. The oil returns by gravity to the oil sump and the air is vented overboard through the drain mast vent.

Carbon Seal

[Refer Figure 49-43](#)

The carbon face seal prevents oil in the bearing area from entering the compressor. The stationary seal gives a leak proof seal between the rotating inner race of the roller bearing and the carbon face seal.

Carbon Seal Drain

A drain system lets any oil that may leak past the carbon seal vent back to the gearbox.

Labyrinth Seal

The Labyrinth Seal is pressed on to a machined shoulder located on the hub of the compressor wheel. The seal uses engine compressor air to give a non-contact air seal between the Labyrinth Seal and the Bearing Area. The air across the seal is vented into the carbon seal drain passage. If oil leaks through the carbon face seal, the Labyrinth Seal prevents oil from entering the compressor.

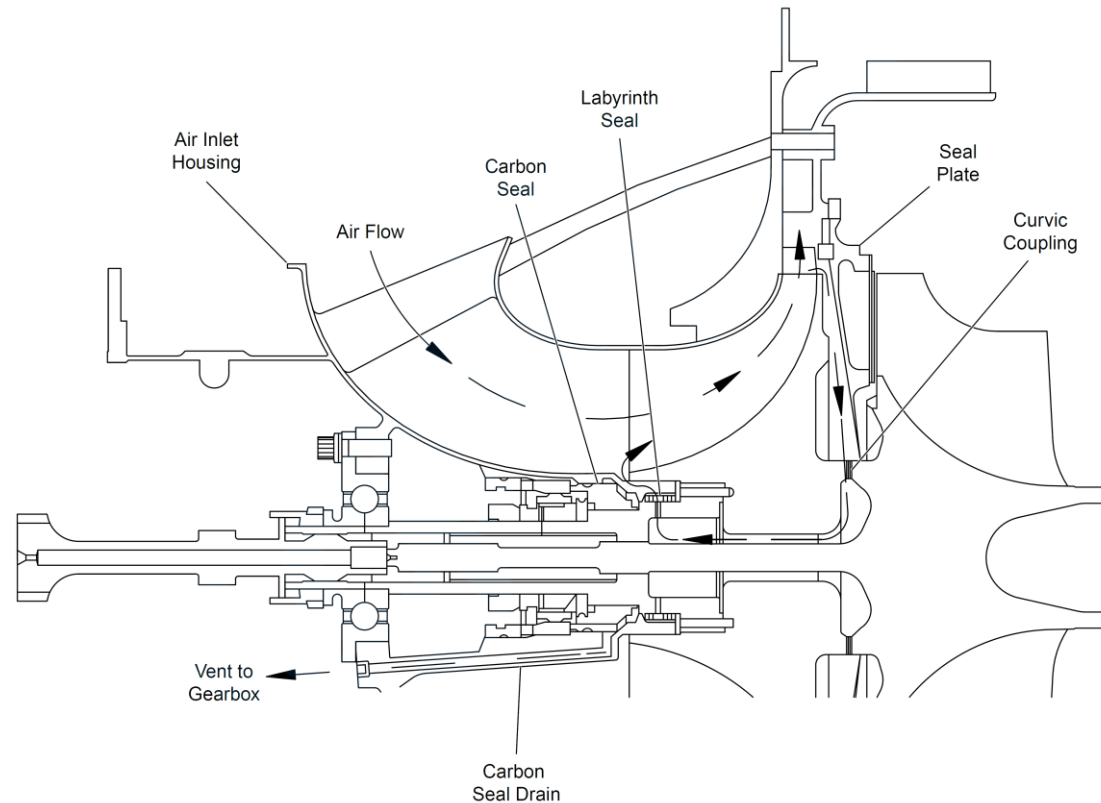


Figure 49-43 Carbon and Labyrinth Seals

Oil Filler Port

[Refer Figure 49-44](#)

The oil system is serviced through the oil filler port. The locking cap is removed and oil is poured into the filler port. The locking cap has a dip stick that is marked ADD and FULL.

Oil Sight Glass

The Oil Sight Glass is located below the Oil Filler Port and gives a visual indication of the oil level in the gearbox oil sump. Proper servicing of the oil system is done by monitoring the sight glass and the dip stick.

Magnetic Drain Plug

A magnetic drain plug is located on the lower, front side of the gearbox housing. When the plug is removed, oil is drained from the oil sump. The drain plug has a magnetic chip detector that attracts ferrous metal particles in the oil. The detector may be removed, inspected, and installed without draining the oil sump. A check valve located inside the drain plug prevents oil from draining when the chip detector is removed.

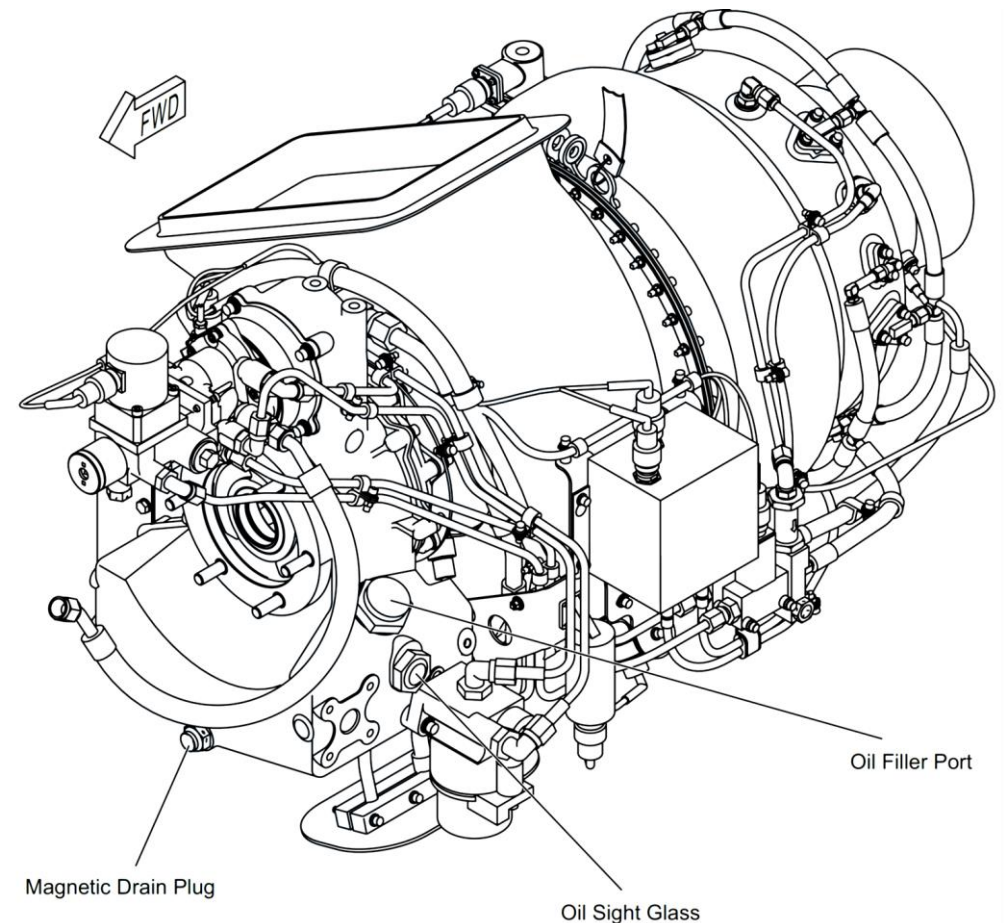


Figure 49-44 Oil Filling, Draining and Checking

Exhaust and Tail Cone Ventilation

Introduction

[Refer Figure 49-45](#)

The APU exhaust system vents the exhaust gases from the APU to the environment, and the ventilation system cools the APU.

General Description

The APU exhaust system consist of an exhaust nozzle and a silenced exhaust duct. This setup also assists in the ventilation of the APU. The ventilation is achieved through a cut out located on the bottom center line on the tailcone.

Detailed Description

The APU exhaust nozzle and the silenced exhaust duct are not connected but arranged to form an ejector system. The exhaust gas is discharged from the duct through an upwards pointing outlet at the aft end of the duct.

NOTE: The exhaust nozzle is removed as part of the APU. The exhaust duct is easily removed after you remove the APU.

The APU and the tailcone are kept cool by the exhaust flow. This exhaust flow, drives the ejector system which cools the APU by ventilation.

Ambient air enters the tailcone through a 4 inch diameter cutout. This is located at the bottom centerline of the tailcone between the firewall and the access door. The cutout is protected by a stainless steel 1/4 inch mesh screen. The mesh screen is located between a riveted titanium doubler and the tailcone skin.

The air is ducted from the cutout by a tube to the area immediately in front of the oil sump to assist in oil cooling. The air then flows around the APU and finally it is ejected into the ejector gas flow.

After shut down, the APU and the surrounding area is cooled by soak back cooling. This is done by convective flow primarily through the normal ventilation system. The principles of convective flow is to have the exhaust

duct outlet at the highest point of the system. This is supplemented by flow around the outside of the exhaust duct, which has a diametrical clearance with the tailcone structure.

NOTE: If you operate the APU with the access doors open, it may result in an oil overheat automatic shutdown.

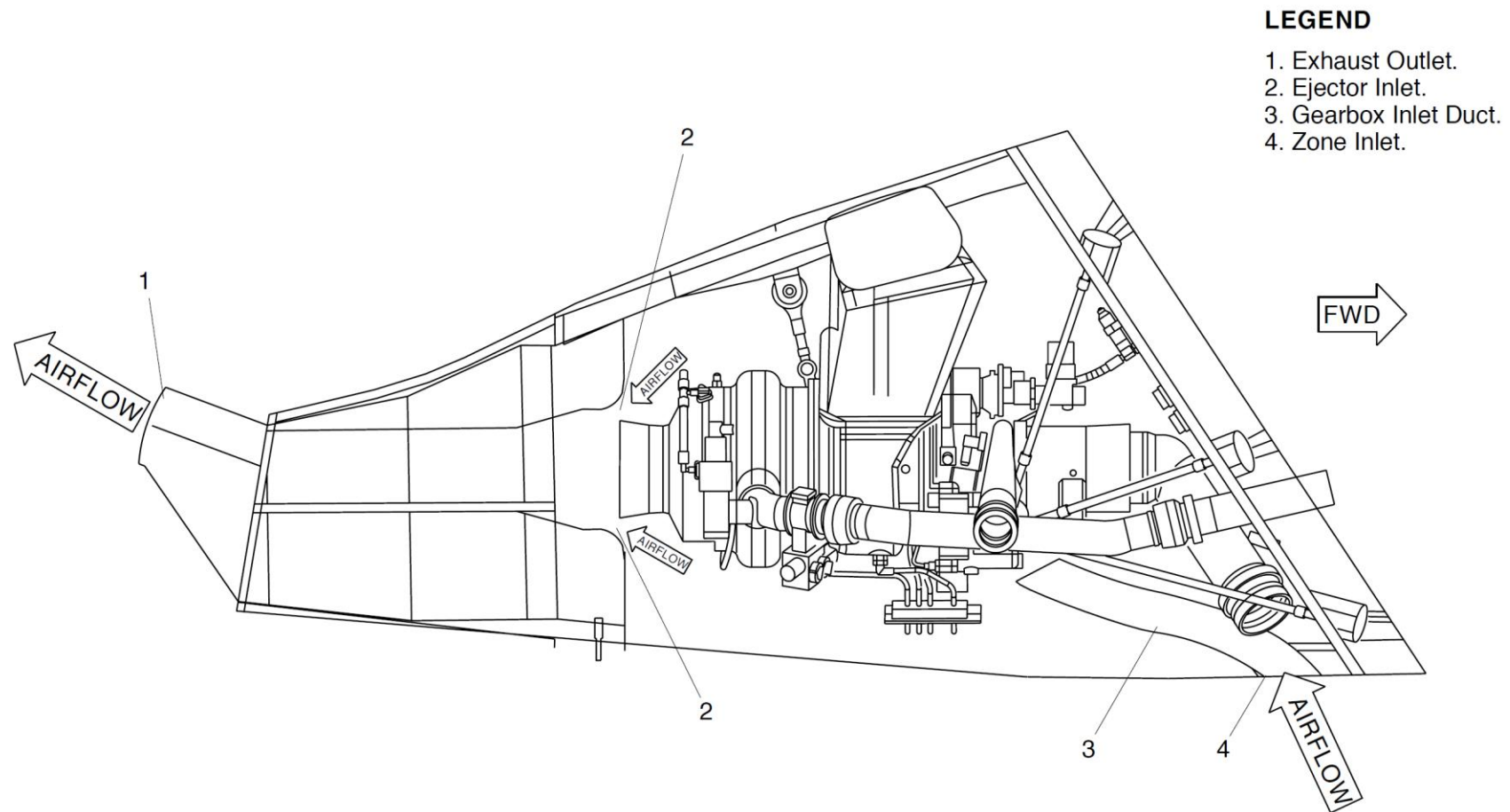


Figure 49-45 APU Exhaust and Tail Cone Ventilation